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Jones

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(54) **MONITORING DATA EXFILTRATION
BASED ON USER STATUS**

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Primary Examiner — Joseph P Hirl

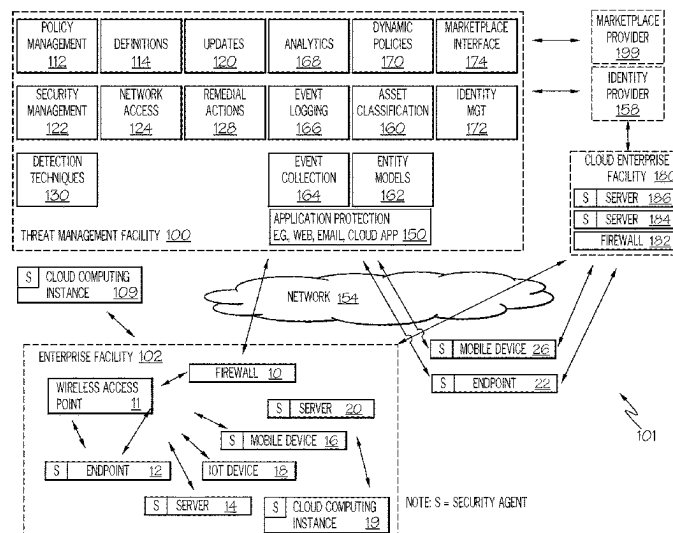
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(57) **ABSTRACT**

A method includes storing an employment status module associated with an entity and associating a plurality of employment statuses with respective security settings for a plurality of users associated with the entity, receiving information indicating a change in an employment status of a user of the plurality of users, determining a change to security settings for the user based on the change in the employment status of the user, applying the change to the security settings for the user in a security management system of the threat management facility, and restricting use of at least one data processing electronic device activity on an electronic device of the user based on the change to the security settings for the user.

20 Claims, 13 Drawing Sheets



(58) **Field of Classification Search**

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7/70491; G01H 2240/101; G10H
2240/105; G11B 20/00144; G11B
20/00804; G02B 27/0093; G05B
2219/32422; G06V 10/235; G06V 40/67;
G06G 21/00; Y02B 60/185; Y02B
90/245; Y02D 10/43; Y10S 705/904

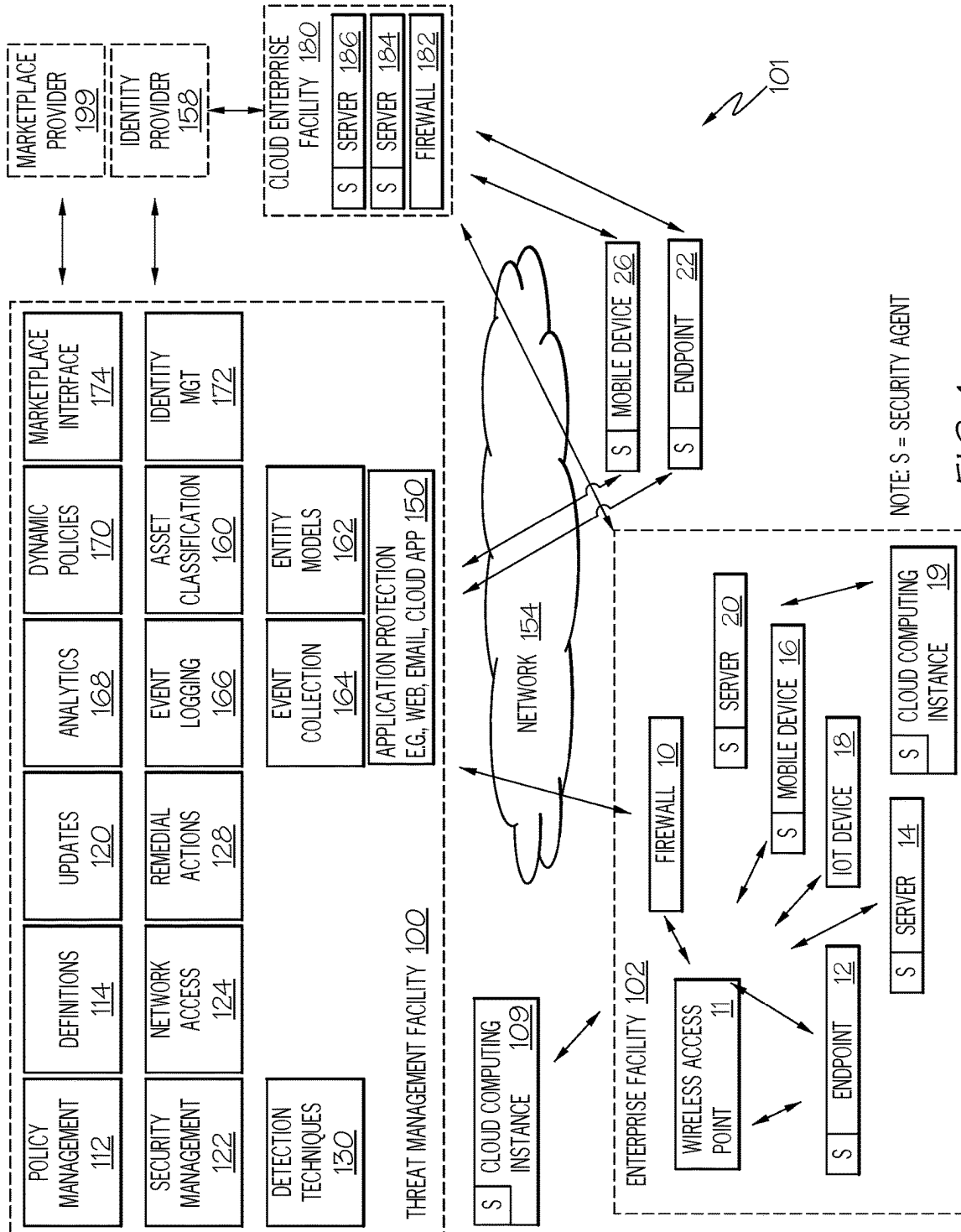
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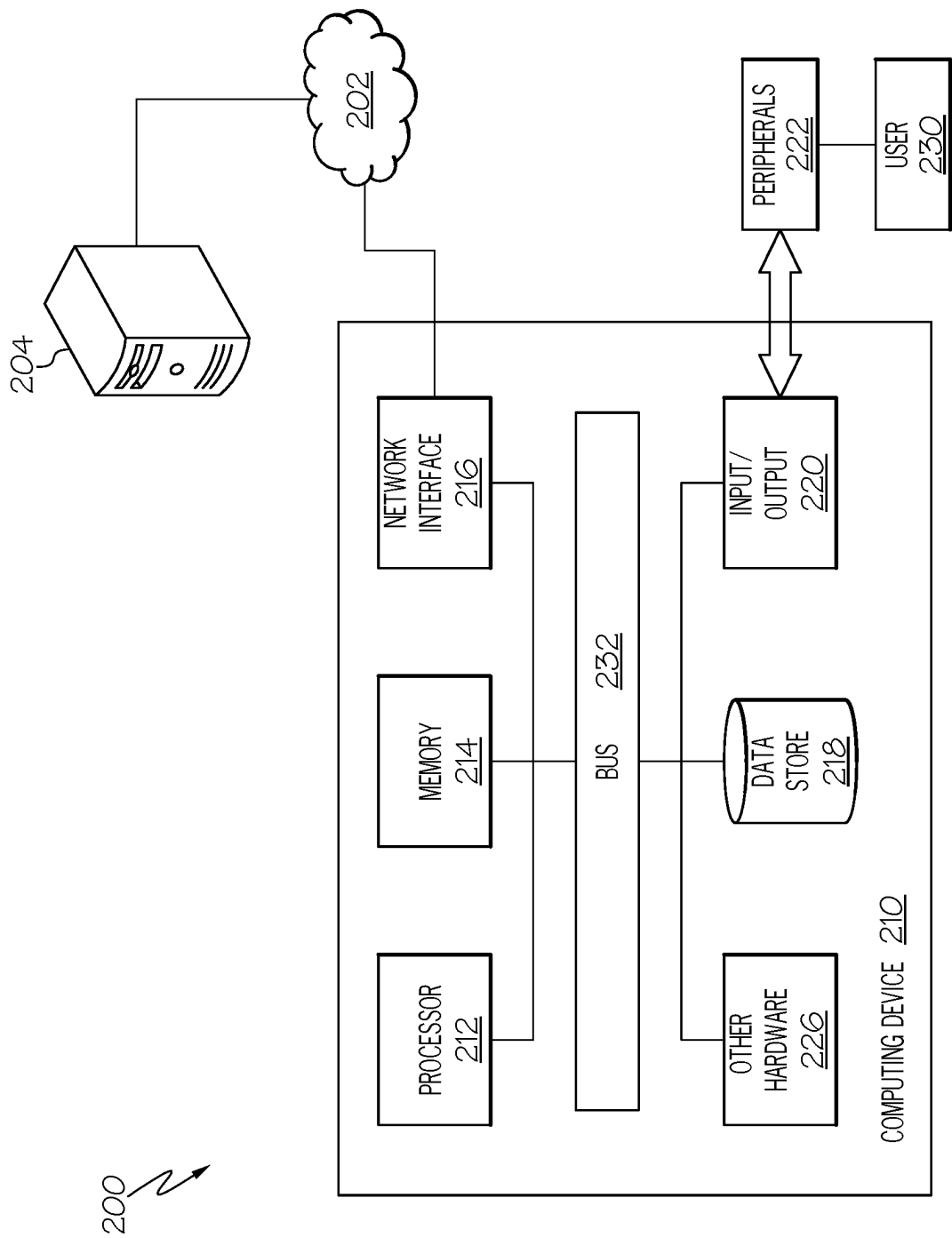


FIG. 2

300 ↗

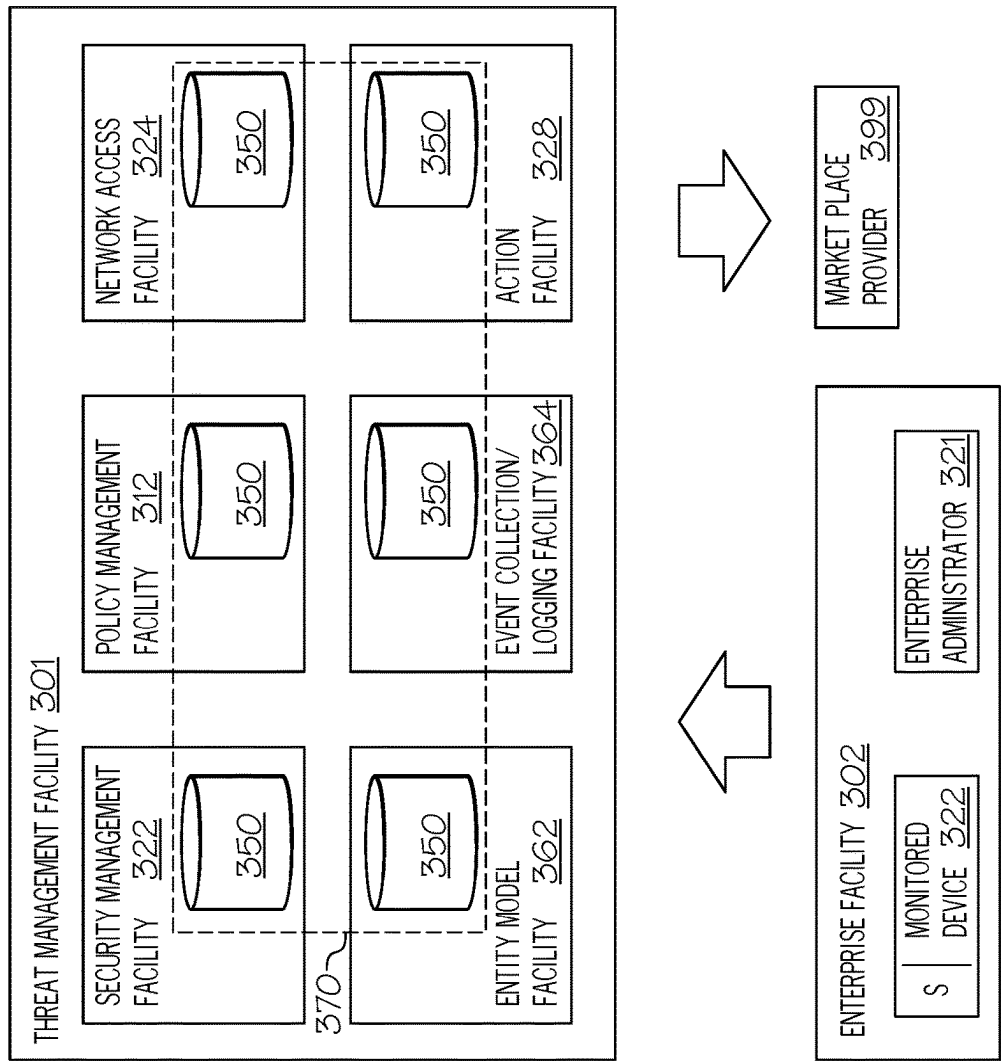


FIG. 3

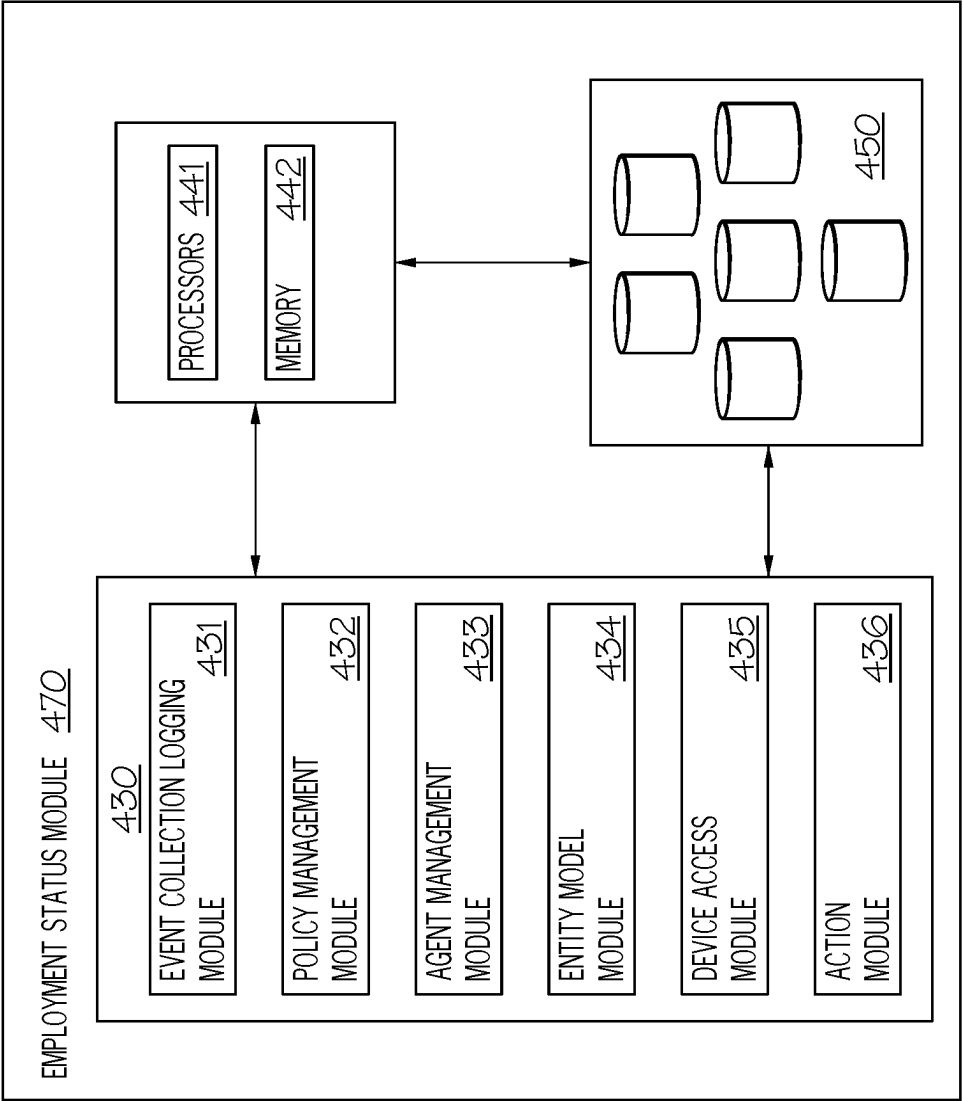


FIG. 4

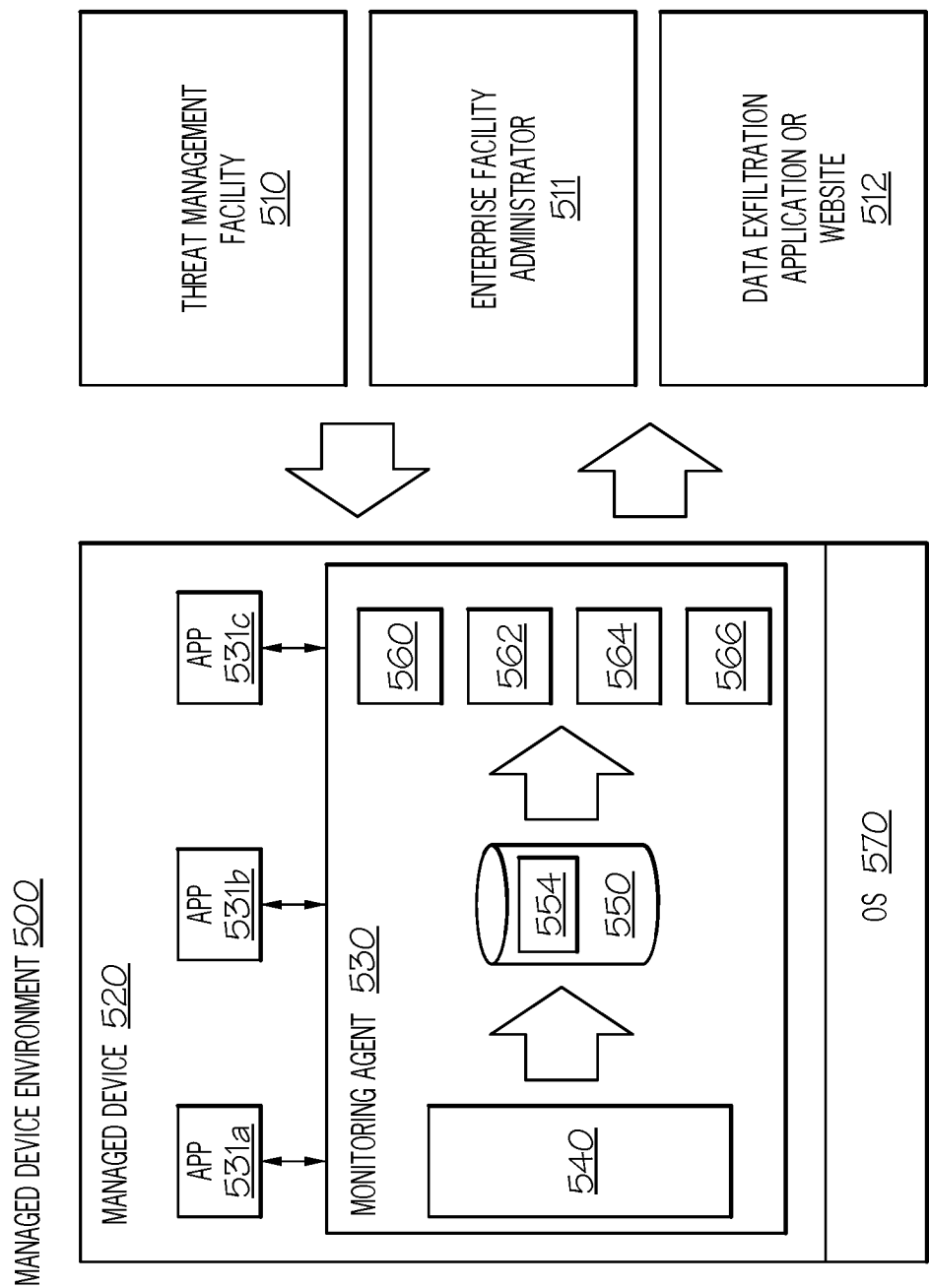


FIG. 5

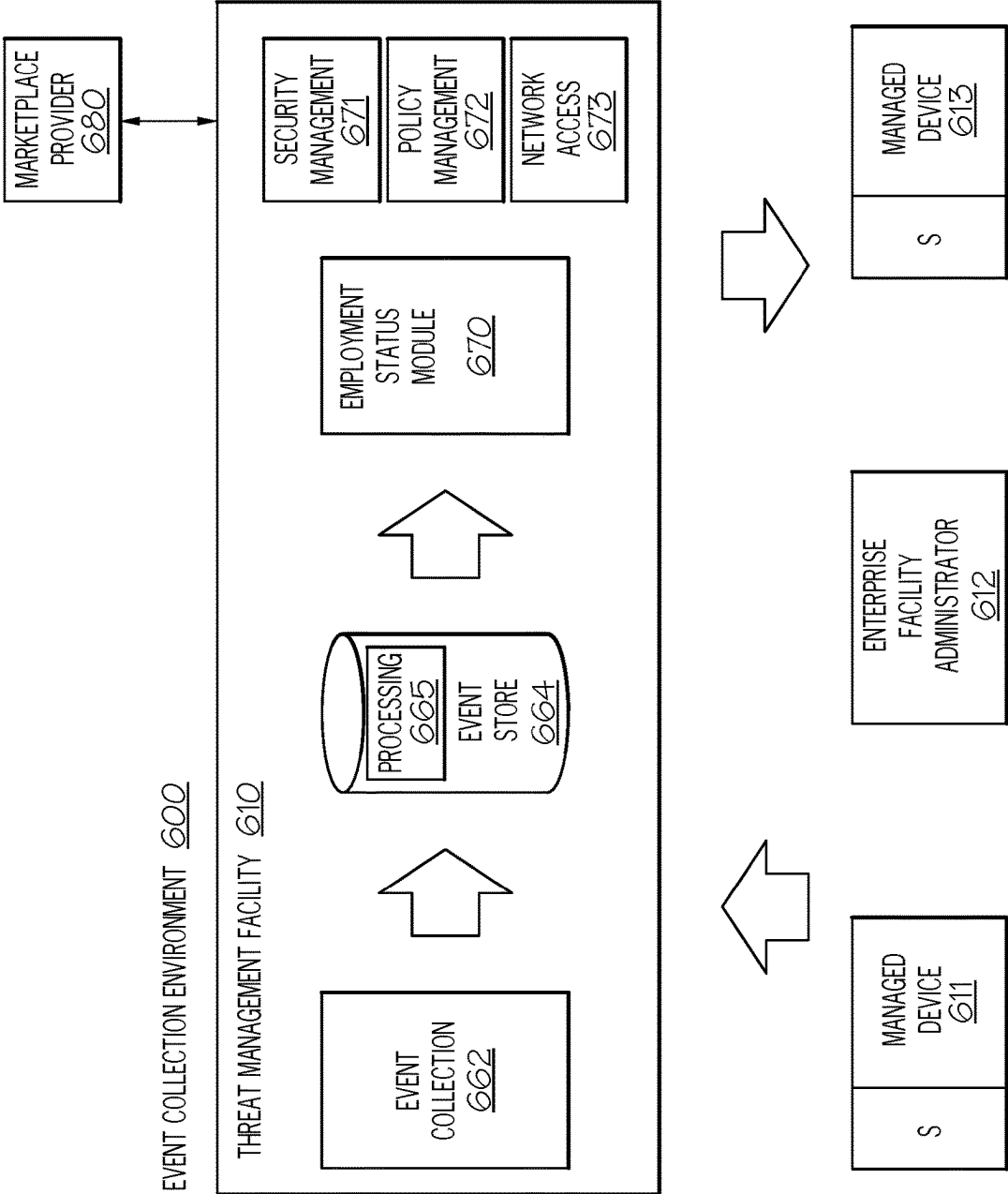


FIG. 6

700 ↗

702 }		704 }		706 }		708 }		710 }		712 }
User	Device(s)	Entity	Employment	Security Setting	User Setting					
User 1	Device 1	1	Employed	E1	Low					
	Device 2									
User 2	Device 1	1	Departing	D1	Med					
User 3	Device 1	1	Departing	D1	Low					
	Device 2					D2	High			
User 4	Device 1	2	Terminated	T1	High					
User 5	Device 1	2	Employed	E1	Low					
User 6	Device 1	2	Employed	E3	High					
	Device 2									

FIG. 7

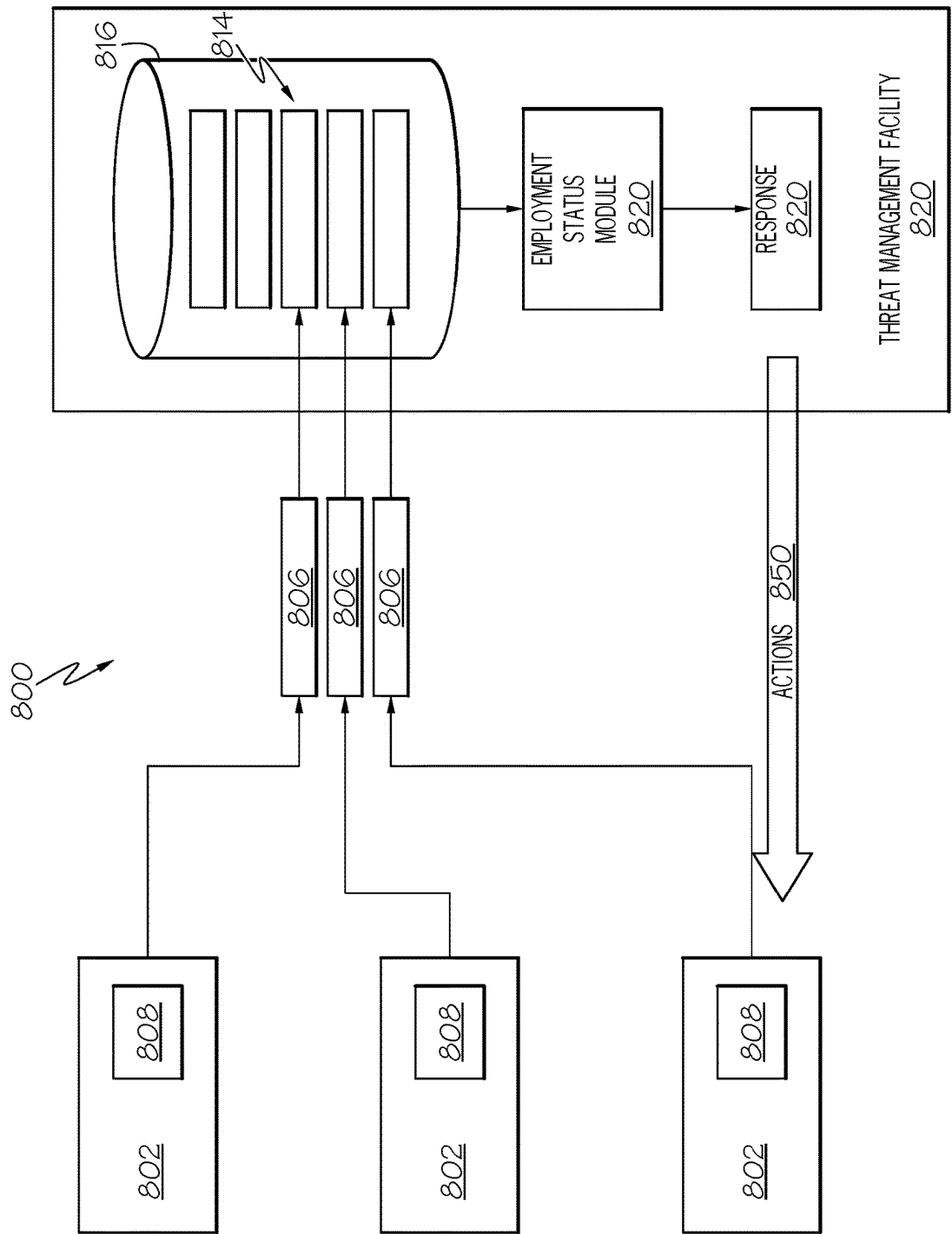


FIG. 8

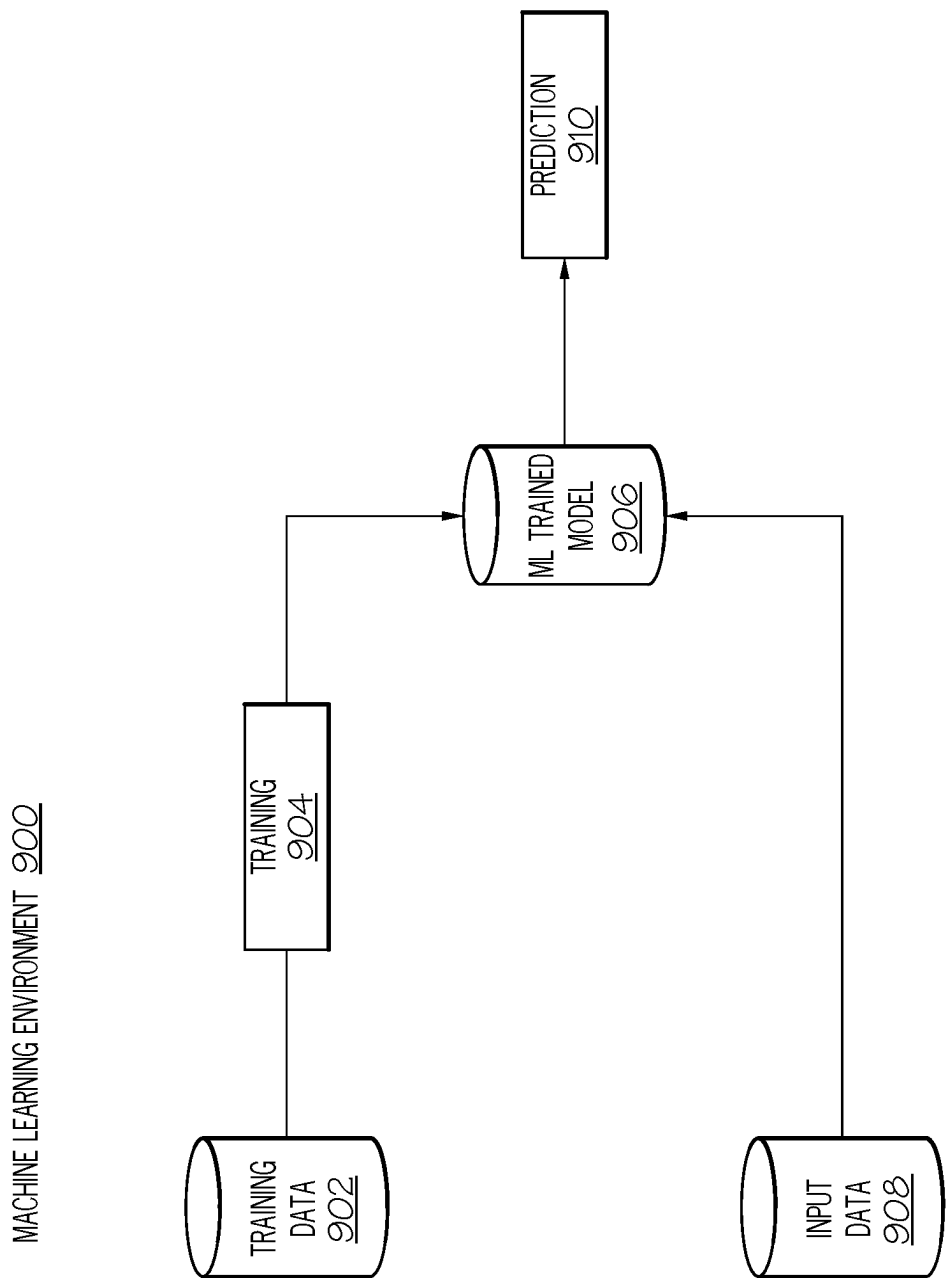


FIG. 9

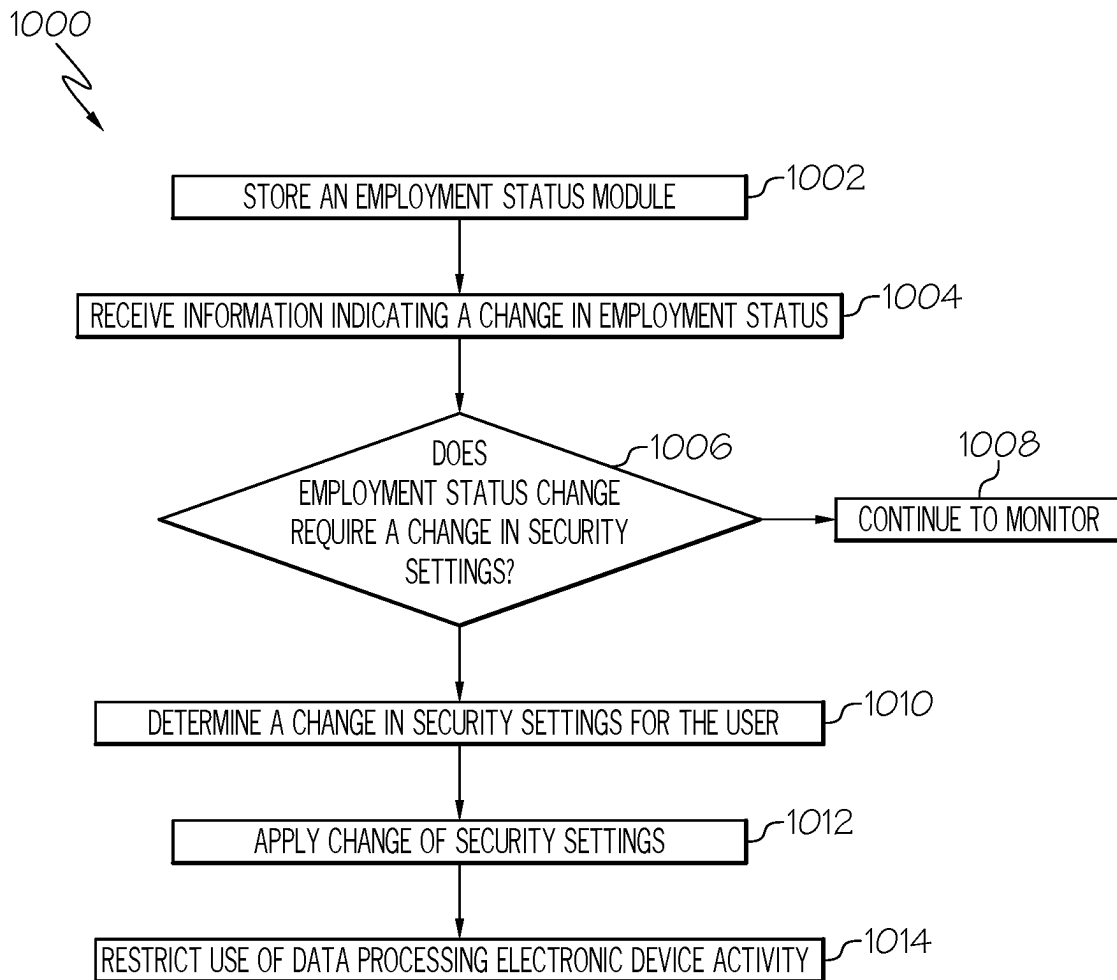


FIG. 10

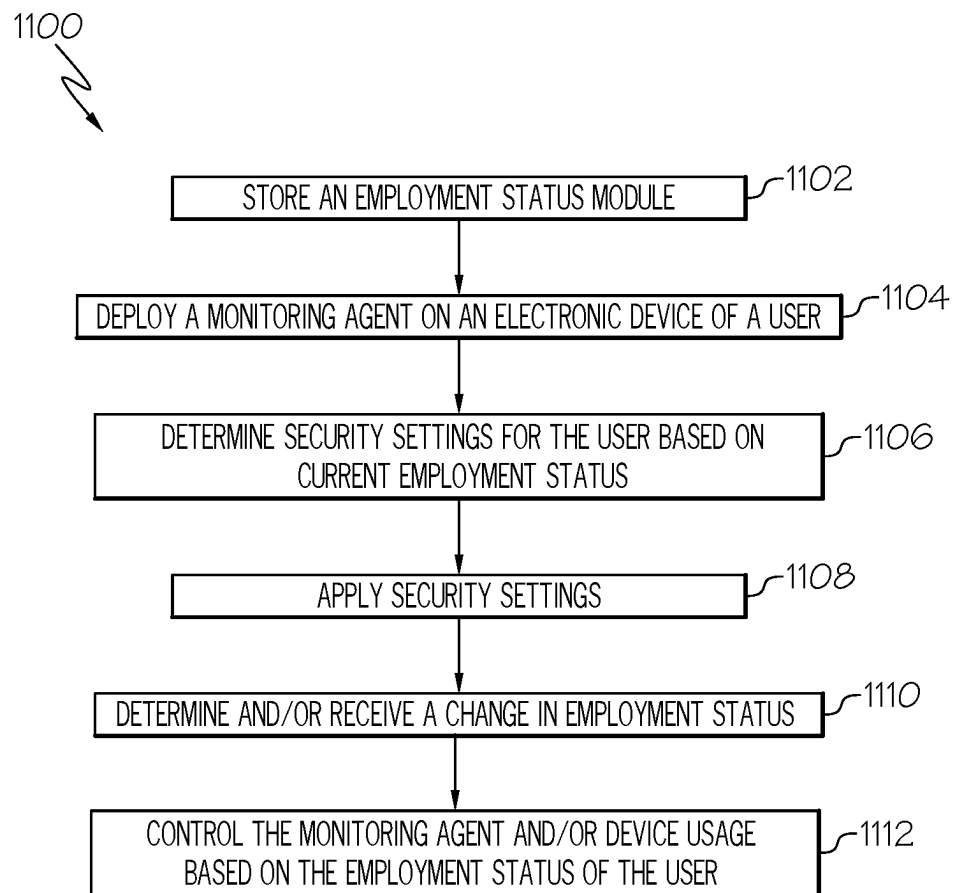


FIG. 11

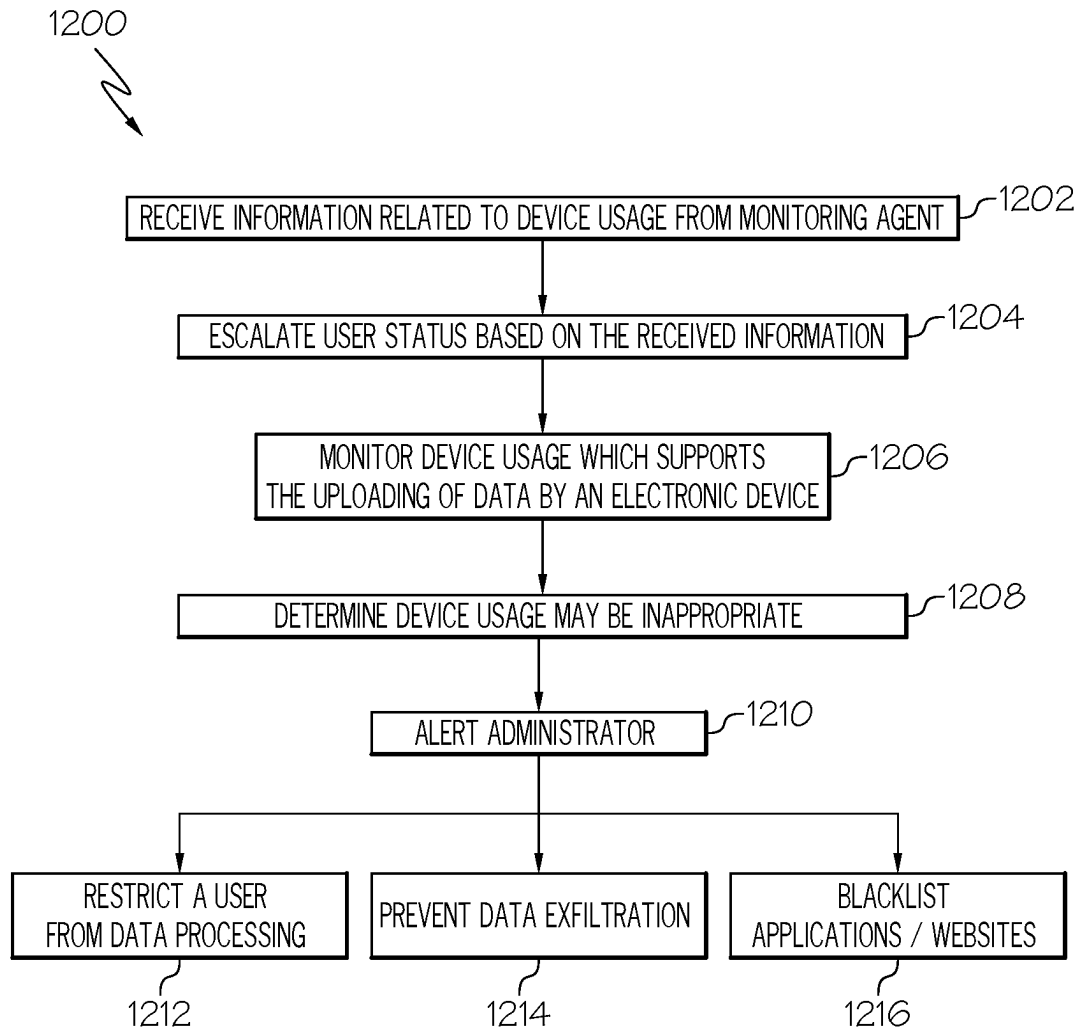


FIG. 12

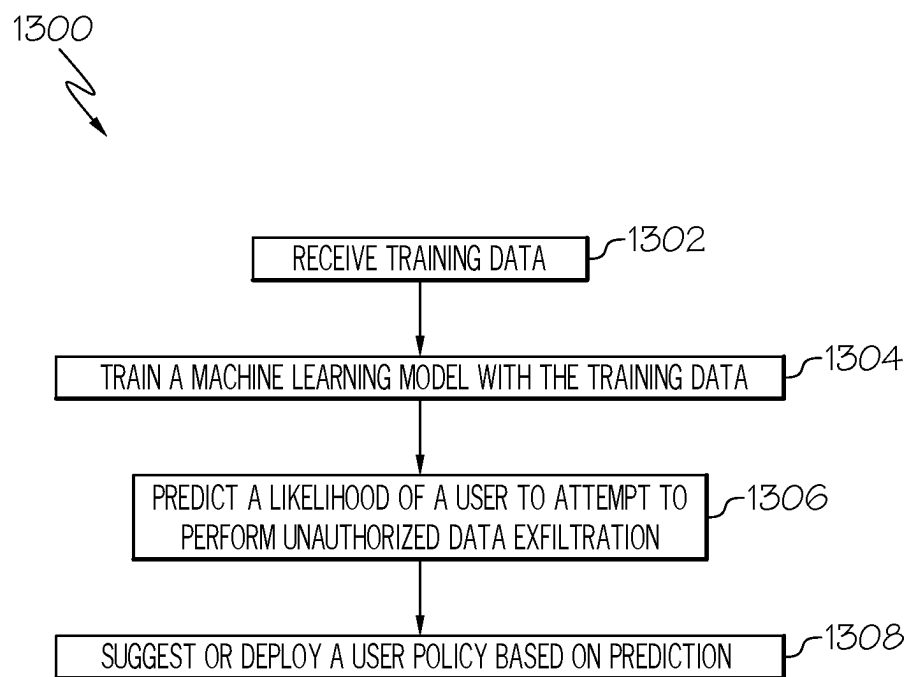


FIG. 13

1

MONITORING DATA EXFILTRATION BASED ON USER STATUS

TECHNICAL FIELD

The present disclosure relates to monitoring user policies and security settings for electronic device usage based on a user status. More specifically, this application describes various systems and/or methods for monitoring data exfiltration based on employment status of a user.

BACKGROUND

When a user is asked to leave an employer, there is a non-trivial likelihood that the user may download one or more files from servers maintained by the employer. The user may download his or her own files, but may also download files that include proprietary, confidential, and/or sensitive information. The user may attempt to upload the downloaded files to one or more data exfiltration websites or applications. Thus, an automated computer system, computer program product and method for monitoring user policies and security settings for electronic device usage of a user based on a user status (e.g. an employment status) would be well received in the art.

SUMMARY

According to various embodiments disclosed herein, a method, and associated computer system and computer program product for monitoring user policies and security settings for electronic device usage based on a user status is provided. A computer system at a threat management facility stores an employment status module associated with at least one entity. The employment status module associates a plurality of employment statuses with respective security settings for a plurality of users associated with the at least one entity. The respective security settings relate to data processing electronic device activity by the plurality of users. One or more processors of the computer system receive information indicating a change in an employment status of a user of the plurality of users. The one or more processors of the computer system determine, using the employment status module, a change to security settings for the user based on the change in the employment status of the user. The one or more processors of the computer system apply the change to the security settings for the user in a security management system of the threat management facility. The method, associated computer system and computer program product then provides for restricting use of at least one data processing electronic device activity on an electronic device of the user based on the change to the security settings for the user.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and further advantages of the foregoing may be better understood by referring to the following description in conjunction with the accompanying drawings, in which like reference numerals indicate like elements and features in the various figures. For clarity, not every element may be labeled in every figure. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the disclosed concepts and features.

FIG. 1 depicts a block diagram of a threat management facility according to an example embodiment.

2

FIG. 2 depicts a computer system according to an example embodiment.

FIG. 3 depicts a system for monitoring user policies and security settings for electronic device usage based on a user status according to an example embodiment.

FIG. 4 depicts an employment status module according to an example embodiment.

FIG. 5 depicts a managed device environment according to an example embodiment.

FIG. 6 depicts an event collection environment according to an example embodiment.

FIG. 7 depicts a policy management table according to an example embodiment.

FIG. 8 depicts a system for monitoring user status and response according to an example embodiment.

FIG. 9 depicts a machine learning environment according to an example embodiment.

FIG. 10 depicts a method, in accordance with an example embodiment, for monitoring user policies and security settings for electronic device usage based on a user status.

FIG. 11 depicts a method, in accordance with an example embodiment, of deploying a monitoring agent.

FIG. 12 depicts a method, in accordance with an example embodiment, for preventing data exfiltration based on user status.

FIG. 13 depicts a method, in accordance with an example embodiment, of predicting a likelihood of a user attempting to perform unauthorized data exfiltration.

DETAILED DESCRIPTION

Reference in the specification to “one embodiment” or “an embodiment” means that a particular, feature, structure or characteristic described in connection with the embodiment is included in at least one embodiment of the teaching. References to a particular embodiment within the specification do not necessarily all refer to the same embodiment.

The disclosed concepts and features are described in more detail with reference to exemplary embodiments thereof as shown in the accompanying drawings. While the various concepts and features are described in conjunction with various embodiments and examples, it is not intended that the concepts and features are limited to such embodiments. On the contrary, the various concepts and features encompass various alternatives, modifications and equivalents, as will be appreciated by those of skill in the art. Those of ordinary skill having access to the concepts described herein will recognize additional implementations, modifications and embodiments, as well as other fields of use, which are within the scope of the present disclosure as described herein.

Recitation of ranges of values herein are not intended to be limiting, referring instead individually to any and all values falling within the range, unless otherwise indicated herein, and each separate value within such a range is incorporated into the specification as if it were individually recited herein. The words “about,” “approximately” or the like, when accompanying a numerical value, are to be construed as indicating a deviation as would be appreciated by one of ordinary skill in the art to operate satisfactorily for an intended purpose. Similarly, words of approximation such as “approximately” or “substantially” when used in reference to physical characteristics, should be understood to contemplate a range of deviations that would be appreciated by one of ordinary skill in the art to operate satisfactorily for a corresponding use, function, purpose, or the like. Ranges of values and/or numeric values are provided herein

3

as examples only, and do not constitute a limitation on the scope of the described embodiments. Where ranges of values are provided, they are also intended to include each value within the range as if set forth individually, unless expressly stated to the contrary. The use of any and all examples, or exemplary language (“e.g.,” “such as,” or the like) provided herein, is intended merely to better illuminate the embodiments and does not pose a limitation on the scope of the embodiments. No language in the specification should be construed as indicating any unclaimed element as essential to the practice of the embodiments.

In the following description, it is understood that terms such as “first,” “second,” “top,” “bottom,” “up,” “down,” and the like, are words of convenience and are not to be construed as limiting terms.

It should also be understood that endpoints, devices, compute instances or the like that are referred to as “within” an enterprise network may also be “associated with” the enterprise network, e.g., where such assets are outside an enterprise gateway but nonetheless managed by or in communication with a threat management facility or other centralized security platform for the enterprise network. Thus, any description referring to an asset within the enterprise network should be understood to contemplate a similar asset associated with the enterprise network regardless of location in a network environment unless a different meaning is explicitly provided or otherwise clear from the context.

Embodiments herein are directed to monitoring and alerting a system administrator to the possibility of data exfiltration based on the employment status of a user or employee. In general, a client device (e.g. an endpoint) includes a security or monitoring agent that monitors data movement on the client device. The monitoring agent described herein may be configured to monitor installed programs and/or websites visited by the user of the client device. In particular, the monitoring agent may monitor websites and applications that support the uploading of data from the client device (e.g., websites, and/or applications that support data exfiltration). Examples of such websites and applications include Microsoft OneDrive, Google Drive, Carbonite, Box, DropBox, and the like. When a user is asked to leave an employer, or there is some other status change to the user, the monitoring agent may be informed of the change in the user’s status. The monitoring agent may then alert an administrator of the client device if and when the user is attempting to upload information which may be classified by the employer as proprietary, confidential and/or containing sensitive information. In some instances, the monitoring agent may be configured to allow the data exfiltration websites prior to a change in the user’s status (e.g. employment status). However, after the change to the user’s status, the same data exfiltration websites may be blocked or prohibited. Various embodiments relating to monitoring user policies and security settings for electronic device usage based on a user status will be described herein.

FIG. 1 depicts a block diagram of a threat management system 101 providing protection against a plurality of threats, such as malware, viruses, spyware, cryptoware, adware, Trojans, spam, intrusion, policy abuse, improper configuration, vulnerabilities, improper access, uncontrolled access, and more according to an example embodiment. The threat management facility 100 may communicate with, coordinate, and control operation of security functionality at different control points, layers, and levels within the system 101. A number of capabilities may be provided by the threat management facility 100, with an overall goal to intelli-

4

gently use the breadth and depth of information that is available about the operation and activity of compute instances and networks as well as a variety of available controls. Another overall goal is to provide protection needed by an organization that is dynamic and able to adapt to changes in compute instances and new threats or unwanted activity. In embodiments, the threat management facility 100 may provide protection from a variety of threats or unwanted activity to an enterprise facility that may include a variety of compute instances in a variety of locations and network configurations.

Just as one example, users of the threat management facility 100 may define and enforce policies that control access to and use of compute instances, networks and data. Administrators may update policies such as by designating authorized users and conditions for use and access. The threat management facility 100 may update and enforce those policies at various levels of control that are available, such as by directing compute instances to control the network traffic that is allowed to traverse firewalls and wireless access points, applications and data available from servers, applications and data permitted to be accessed by endpoints, and network resources and data permitted to be run and used by endpoints. The threat management facility 100 may provide many different services, and policy management may be offered as one of the services.

Turning to a description of certain capabilities and components of the threat management system 101, an exemplary enterprise facility 102 may be or may include any networked computer-based infrastructure. For example, the enterprise facility 102 may be corporate, commercial, organizational, educational, governmental, or the like. As home networks get more complicated, and include more compute instances at home and in the cloud, an enterprise facility 102 may also or instead include a personal network such as a home or a group of homes. The enterprise facility’s 102 computer network may be distributed amongst a plurality of physical premises such as buildings on a campus, and located in one or in a plurality of geographical locations. The configuration of the enterprise facility as shown is merely exemplary, and it will be understood that there may be any number of compute instances, less or more of each type of compute instances, and other types of compute instances. As shown, the exemplary enterprise facility includes a firewall 10, a wireless access point 11, an endpoint 12, a server 14, a mobile device 16, an appliance or TOT device 18, a cloud computing instance 19, and a server 20. Again, the compute instances 10-20 depicted are exemplary, and there may be any number or types of compute instances 10-20 in a given enterprise facility. For example, in addition to the elements depicted in the enterprise facility 102, there may be one or more gateways, bridges, wired networks, wireless networks, virtual private networks, other compute instances, and so on.

The threat management facility 100 may include certain facilities, such as a policy management facility 112, security management facility 122, update facility 120, definitions facility 114, network access rules facility 124, remedial action facility 128, detection techniques facility 130, application protection facility 150, asset classification facility 160, entity model facility 162, event collection facility 164, event logging facility 166, analytics facility 168, dynamic policies facility 170, identity management facility 172, and marketplace management facility 174, as well as other facilities. For example, there may be a testing facility, a threat research facility, and other facilities. It should be understood that the threat management facility 100 may be implemented in whole or in part on a number of different

compute instances, with some parts of the threat management facility on different compute instances in different locations. For example, some or all of one or more of the various facilities **100**, **112-174** may be provided as part of a security agent **S** that is included in software running on a compute instance **10-26** within the enterprise facility. Some or all of one or more of the facilities **100**, **112-174** may be provided on the same physical hardware or logical resource as a gateway, such as a firewall **10**, or wireless access point **11**. Some or all of one or more of the facilities may be provided on one or more cloud servers that are operated by the enterprise or by a security service provider, such as the cloud computing instance **109**.

In embodiments, a marketplace provider **199** may make available one or more additional facilities to the enterprise facility **102** via the threat management facility **100**. The marketplace provider may communicate with the threat management facility **100** via the marketplace interface facility **174** to provide additional functionality or capabilities to the threat management facility **100** and compute instances **10-26**. A marketplace provider **199** may be selected from a number of providers in a marketplace of providers that are available for integration or collaboration via the marketplace interface facility **174**. A given marketplace provider **199** may use the marketplace interface facility **174** even if not engaged or enabled from or in a marketplace. As non-limiting examples, the marketplace provider **199** may be a third-party information provider, such as a physical security event provider; the marketplace provider **199** may be a system provider, such as a human resources system provider or a fraud detection system provider; the marketplace provider **199** may be a specialized analytics provider; and so on. The marketplace provider **199**, with appropriate permissions and authorization, may receive and send events, observations, inferences, controls, convictions, policy violations, or other information to the threat management facility. For example, the marketplace provider **199** may subscribe to and receive certain events, and in response, based on the received events and other events available to the marketplace provider **199**, send inferences to the marketplace interface, and in turn to the analytics facility **168**, which in turn may be used by the security management facility **122**.

The identity provider **158** may be any remote identity management system or the like configured to communicate with an identity management facility **172**, e.g., to confirm identity of a user as well as provide or receive other information about users that may be useful to protect against threats. In general, the identity provider may be any system or entity that creates, maintains, and manages identity information for principals while providing authentication services to relying party applications, e.g., within a federation or distributed network. The identity provider may, for example, offer user authentication as a service, where other applications, such as web applications, outsource the user authentication step to a trusted identity provider.

In embodiments, the identity provider **158** may provide user identity information, such as multi-factor authentication, to a SaaS application. Centralized identity providers such as Microsoft Azure, may be used by an enterprise facility instead of maintaining separate identity information for each application or group of applications, and as a centralized point for integrating multifactor authentication. In embodiments, the identity management facility **172** may communicate hygiene, or security risk information, to the identity provider **158**. The identity management facility **172** may determine a risk score for a user based on the events, observations, and inferences about that user and the compute

instances associated with the user. If a user is perceived as risky, the identity management facility **172** can inform the identity provider **158**, and the identity provider **158** may take steps to address the potential risk, such as to confirm the identity of the user, confirm that the user has approved the SaaS application access, remediate the user's system, or such other steps as may be useful.

In embodiments, threat protection provided by the threat management facility **100** may extend beyond the network boundaries of the enterprise facility **102** to include clients (or client facilities) such as an endpoint **22** outside the enterprise facility **102**, a mobile device **26**, a cloud computing instance **109**, or any other devices, services or the like that use network connectivity not directly associated with or controlled by the enterprise facility **102**, such as a mobile network, a public cloud network, or a wireless network at a hotel or coffee shop. While threats may come from a variety of sources, such as from network threats, physical proximity threats, secondary location threats, the compute instances **10-26** may be protected from threats even when a compute instance **10-26** is not connected to the enterprise facility **102** network, such as when compute instances **22**, **26** use a network that is outside of the enterprise facility **102** and separated from the enterprise facility **102**, e.g., by a gateway, a public network, and so forth.

In some implementations, compute instances **10-26** may communicate with a cloud enterprise facility **280**. The cloud enterprise facility may include one or more cloud applications, such as a SaaS application, which is used by but not operated by the enterprise facility **102**. Exemplary commercially available SaaS applications include Salesforce, Amazon Web Services (AWS) applications, Google Apps applications, Microsoft Office **365** applications and so on. A given SaaS application may communicate with an identity provider **158** to verify user identity consistent with the requirements of the enterprise facility **102**. The compute instances **10-26** may communicate with an unprotected server (not shown) such as a web site or a third-party application through an internetwork **154** such as the Internet or any other public network, private network or combination of these.

The cloud enterprise facility **280** may include servers **284**, **286**, and a firewall **282**. The servers **284**, **286** on the cloud enterprise facility **280** may run one or more enterprise or cloud applications, such as SaaS applications, and make them available to the enterprise facilities **102** compute instances **10-26**. It should be understood that there may be any number of servers **284**, **286** and firewalls **282**, as well as other compute instances in a given cloud enterprise facility **280**. It also should be understood that a given enterprise facility may use both SaaS applications and cloud enterprise facilities **280**, or, for example, a SaaS application may be deployed on a cloud enterprise facility **280**.

In embodiments, aspects of the threat management facility **100** may be provided as a stand-alone solution. In other embodiments, aspects of the threat management facility **100** may be integrated into a third-party product. An application programming interface (e.g. a source code interface) may be provided such that aspects of the threat management facility **100** may be integrated into or used by or with other applications. For instance, the threat management facility **100** may be stand-alone in that it provides direct threat protection to an enterprise or computer resource, where protection is subscribed to directly **100**. Alternatively, the threat management facility may offer protection indirectly, through a third-party product, where an enterprise may subscribe to services through the third-party product, and

threat protection to the enterprise may be provided by the threat management facility **100** through the third-party product.

The security management facility **122** may provide protection from a variety of threats by providing, as non-limiting examples, endpoint security and control, email security and control, web security and control, reputation-based filtering, machine learning classification, control of unauthorized users, control of guest and non-compliant computers, and more.

The security management facility **122** may provide malicious code protection to a compute instance. The security management facility **122** may include functionality to scan applications, files, and data for malicious code, remove or quarantine applications and files, prevent certain actions, perform remedial actions, as well as other security measures. Scanning may use any of a variety of techniques, including without limitation signatures, identities, classifiers, and other suitable scanning techniques. In embodiments, the scanning may include scanning some or all files on a periodic basis, scanning an application when the application is executed, scanning data transmitted to or from a device, scanning in response to predetermined actions or combinations of actions, and so forth. The scanning of applications, files, and data may be performed to detect known or unknown malicious code or unwanted applications. Aspects of the malicious code protection may be provided, for example, in the security agent of an endpoint **12**, in a wireless access point **11** or firewall **10**, as part of application protection **150** provided by the cloud, and so on.

In an embodiment, the security management facility **122** may provide for email security and control, for example to target spam, viruses, spyware and phishing, to control email content, and the like. Email security and control may protect against inbound and outbound threats, protect email infrastructure, prevent data leakage, provide spam filtering, and more. Aspects of the email security and control may be provided, for example, in the security agent of an endpoint **12**, in a wireless access point **11** or firewall **10**, as part of application protection **150** provided by the cloud, and so on.

In an embodiment, security management facility **122** may provide for web security and control, for example, to detect or block viruses, spyware, malware, unwanted applications, help control web browsing, and the like, which may provide comprehensive web access control enabling safe, productive web browsing. Web security and control may provide Internet use policies, reporting on suspect compute instances, security and content filtering, active monitoring of network traffic, URI filtering, and the like. Aspects of the web security and control may be provided, for example, in the security agent of an endpoint **12**, in a wireless access point **11** or firewall **10**, as part of application protection **150** provided by the cloud, and so on.

In an embodiment, the security management facility **122** may provide for network access control, which generally controls access to and use of network connections. Network control may stop unauthorized, guest, or non-compliant systems from accessing networks, and may control network traffic that is not otherwise controlled at the client level. In addition, network access control may control access to virtual private networks (VPN), where VPNs may, for example, include communications networks tunneled through other networks and establishing logical connections acting as virtual networks. In embodiments, a VPN may be treated in the same manner as a physical network. Aspects of network access control may be provided, for example, in the security agent of an endpoint **12**, in a wireless access

point **11** or firewall **10**, as part of application protection **150** provided by the cloud, e.g., from the threat management facility **100** or other network resource(s).

In an embodiment, the security management facility **122** may provide for host intrusion prevention through behavioral monitoring and/or runtime monitoring, which may guard against unknown threats by analyzing application behavior before or as an application runs. This may include monitoring code behavior, application programming interface calls made to libraries or to the operating system, or otherwise monitoring application activities. Monitored activities may include, for example, reading and writing to memory, reading and writing to disk, network communication, process interaction, and so on. Behavior and runtime monitoring may intervene if code is deemed to be acting in a manner that is suspicious or malicious. Aspects of behavior and runtime monitoring may be provided, for example, in the security agent of an endpoint **12**, in a wireless access point **11** or firewall **10**, as part of application protection **150** provided by the cloud, and so on.

In an embodiment, the security management facility **122** may provide for reputation filtering, which may target or identify sources of known malware. For instance, reputation filtering may include lists of URIs of known sources of malware or known suspicious IP addresses, code authors, code signers, or domains, that when detected may invoke an action by the threat management facility **100**. Based on reputation, potential threat sources may be blocked, quarantined, restricted, monitored, or some combination of these, before an exchange of data can be made. Aspects of reputation filtering may be provided, for example, in the security agent of an endpoint **12**, in a wireless access point **11** or firewall **10**, as part of application protection **150** provided by the cloud, and so on. In embodiments, some reputation information may be stored on a compute instance **10-26**, and other reputation data available through cloud lookups to an application protection lookup database, such as may be provided by application protection **150**.

In embodiments, information may be sent from the enterprise facility **102** to a third party, such as a security vendor, or the like, which may lead to improved performance of the threat management facility **100**. In general, feedback may be useful for any aspect of threat detection. For example, the types, times, and number of virus interactions that an enterprise facility **102** experiences may provide useful information for the preventions of future virus threats. Feedback may also be associated with behaviors of individuals within the enterprise, such as being associated with most common violations of policy, network access, unauthorized application loading, unauthorized external device use, and the like. In embodiments, feedback may enable the evaluation or profiling of client actions that are violations of policy that may provide a predictive model for the improvement of enterprise policies.

An update management facility **120** may provide control over when updates are performed. The updates may be automatically transmitted, manually transmitted, or some combination of these. Updates may include software, definitions, reputations or other code or data that may be useful to the various facilities. For example, the update facility **120** may manage receiving updates from a provider, distribution of updates to enterprise facility **102** networks and compute instances, or the like. In embodiments, updates may be provided to the enterprise facility's **102** network, where one or more compute instances on the enterprise facility's **102** network may distribute updates to other compute instances.

The threat management facility **100** may include a policy management facility **112** that manages rules or policies for the enterprise facility **102**. Exemplary rules include access permissions associated with networks, applications, compute instances, users, content, data, and the like. The policy management facility **112** may use a database, a text file, other data store, or a combination to store policies. In an embodiment, a policy database may include a block list, a black list, an allowed list, a white list, and more. As a few non-limiting examples, policies may include a list of enterprise facility **102** external network locations/applications that may or may not be accessed by compute instances, a list of types/classifications of network locations or applications that may or may not be accessed by compute instances, and contextual rules to evaluate whether the lists apply. For example, there may be a rule that does not permit access to sporting websites. When a website is requested by the client facility, a security management facility **122** may access the rules within a policy facility to determine if the requested access is related to a sporting website.

The policy management facility **112** may include access rules and policies that are distributed to maintain control of access by the compute instances **10-26** to network resources. Exemplary policies may be defined for an enterprise facility, application type, subset of application capabilities, organization hierarchy, compute instance type, user type, network location, time of day, connection type, or any other suitable definition. Policies may be maintained through the threat management facility **100**, in association with a third party, or the like. For example, a policy may restrict instant messaging (IM) activity by limiting such activity to support personnel when communicating with customers. More generally, this may allow communication for departments as necessary or helpful for department functions, but may otherwise preserve network bandwidth for other activities by restricting the use of IM to personnel that need access for a specific purpose. In an embodiment, the policy management facility **112** may be a stand-alone application, may be part of the network server facility **142**, may be part of the enterprise facility **102** network, may be part of the client facility, or any suitable combination of these.

The policy management facility **112** may include dynamic policies that use contextual or other information to make security decisions. As described herein, the dynamic policies facility **170** may generate policies dynamically based on observations and inferences made by the analytics facility. The dynamic policies generated by the dynamic policy facility **170** may be provided by the policy management facility **112** to the security management facility **122** for enforcement.

In embodiments, the threat management facility **100** may provide configuration management as an aspect of the policy management facility **112**, the security management facility **122**, or some combination. Configuration management may define acceptable or required configurations for the compute instances **10-26**, applications, operating systems, hardware, or other assets, and manage changes to these configurations. Assessment of a configuration may be made against standard configuration policies, detection of configuration changes, remediation of improper configurations, application of new configurations, and so on. An enterprise facility may have a set of standard configuration rules and policies for particular compute instances which may represent a desired state of the compute instance. For example, on a given compute instance **12**, **14**, **18**, a version of a client firewall may be required to be running and installed. If the required version is installed but in a disabled state, the policy violation may prevent

access to data or network resources. A remediation may be to enable the firewall. In another example, a configuration policy may disallow the use of USB disks, and policy management **112** may require a configuration that turns off USB drive access via a registry key of a compute instance. Aspects of configuration management may be provided, for example, in the security agent of an endpoint **12**, in a wireless access point **11** or firewall **10**, as part of application protection **150** provided by the cloud, or any combination of these.

In embodiments, the threat management facility **100** may also provide for the isolation or removal of certain applications that are not desired or may interfere with the operation of a compute instance **10-26** or the threat management facility **100**, even if such application is not malware per se. The operation of such products may be considered a configuration violation. The removal of such products may be initiated automatically whenever such products are detected, or access to data and network resources may be restricted when they are installed and running. In the case where such applications are services which are provided indirectly through a third-party product, the applicable application or processes may be suspended until action is taken to remove or disable the third-party product.

The policy management facility **112** may also require update management (e.g., as provided by the update facility **120**). Update management for the security facility **122** and policy management facility **112** may be provided directly by the threat management facility **100**, or, for example, by a hosted system. In embodiments, the threat management facility **100** may also provide for patch management, where a patch may be an update to an operating system, an application, a system tool, or the like, where one of the reasons for the patch is to reduce vulnerability to threats.

In embodiments, the security facility **122** and policy management facility **112** may push information to the enterprise facility **102** network and/or the compute instances **10-26**, the enterprise facility **102** network and/or compute instances **10-26** may pull information from the security facility **122** and policy management facility **112**, or there may be a combination of pushing and pulling of information. For example, the enterprise facility **102** network and/or compute instances **10-26** may pull update information from the security facility **122** and policy management facility **112** via the update facility **120**, an update request may be based on a time period, by a certain time, by a date, on demand, or the like. In another example, the security facility **122** and policy management facility **112** may push the information to the enterprise facility's **102** network and/or compute instances **10-26** by providing notification that there are updates available for download and/or transmitting the information. In an embodiment, the policy management facility **112** and the security facility **122** may work in concert with the update management facility **120** to provide information to the enterprise facility's **102** network and/or compute instances **10-26**. In various embodiments, policy updates, security updates and other updates may be provided by the same or different modules, which may be the same or separate from a security agent running on one of the compute instances **10-26**.

As threats are identified and characterized, the definition facility **114** of the threat management facility **100** may manage definitions used to detect and remediate threats. For example, identity definitions may be used for scanning files, applications, data streams, etc. for the determination of malicious code. Identity definitions may include instructions and data that can be parsed and acted upon for recognizing

11

features of known or potentially malicious code. Definitions also may include, for example, code or data to be used in a classifier, such as a neural network or other classifier that may be trained using machine learning. Updated code or data may be used by the classifier to classify threats. In embodiments, the threat management facility **100** and the compute instances **10-26** may be provided with new definitions periodically to include most recent threats. Updating of definitions may be managed by the update facility **120**, and may be performed upon request from one of the compute instances **10-26**, upon a push, or some combination. Updates may be performed upon a time period, on demand from a device **10-26**, upon determination of an important new definition or a number of definitions, and so on.

A threat research facility (not shown) may provide a continuously ongoing effort to maintain the threat protection capabilities of the threat management facility **100** in light of continuous generation of new or evolved forms of malware. Threat research may be provided by researchers and analysts working on known threats, in the form of policies, definitions, remedial actions, and so on.

The security management facility **122** may scan an outgoing file and verify that the outgoing file is permitted to be transmitted according to policies. By checking outgoing files, the security management facility **122** may be able to discover threats that were not detected on one of the compute instances **10-26**, or policy violation, such as transmission of information that should not be communicated unencrypted.

The threat management facility **100** may control access to the enterprise facility **102** networks. A network access facility **124** may restrict access to certain applications, networks, files, printers, servers, databases, and so on. In addition, the network access facility **124** may restrict user access under certain conditions, such as the user's location, usage history, need to know, job position, connection type, time of day, method of authentication, client-system configuration, or the like. Network access policies may be provided by the policy management facility **112**, and may be developed by the enterprise facility **102**, or pre-packaged by a supplier. Network access facility **124** may determine if a given compute instance **10-22** should be granted access to a requested network location, e.g., inside or outside of the enterprise facility **102**. Network access facility **124** may determine if a compute instance **22, 26** such as a device outside the enterprise facility **102** may access the enterprise facility **102**. For example, in some cases, the policies may require that when certain policy violations are detected, certain network access is denied. The network access facility **124** may communicate remedial actions that are necessary or helpful to bring a device back into compliance with policy as described below with respect to the remedial action facility **128**. Aspects of the network access facility **124** may be provided, for example, in the security agent of the endpoint **12**, in a wireless access point **11**, in a firewall **10**, as part of application protection **150** provided by the cloud, and so on.

In an embodiment, the network access facility **124** may have access to policies that include one or more of a block list, a black list, an allowed list, a white list, an unacceptable network site database, an acceptable network site database, a network site reputation database, or the like of network access locations that may or may not be accessed by the client facility. Additionally, the network access facility **124** may use rule evaluation to parse network access requests and apply policies. The network access rule facility **124** may have a generic set of policies for all compute instances, such as denying access to certain types of websites, controlling

12

instant messenger accesses, or the like. Rule evaluation may include regular expression rule evaluation, or other rule evaluation method(s) for interpreting the network access request and comparing the interpretation to established rules for network access. Classifiers may be used, such as neural network classifiers or other classifiers that may be trained by machine learning.

The threat management facility **100** may include an asset classification facility **160**. The asset classification facility will discover the assets present in the enterprise facility **102**. A compute instance such as any of the compute instances **10-26** described herein may be characterized as a stack of assets. The one level asset is an item of physical hardware. The compute instance may be, or may be implemented on physical hardware, and may have or may not have a hypervisor, or may be an asset managed by a hypervisor. The compute instance may have an operating system (e.g., Windows, MacOS, Linux, Android, iOS). The compute instance may have one or more layers of containers. The compute instance may have one or more applications, which may be native applications, e.g., for a physical asset or virtual machine, or running in containers within a computing environment on a physical asset or virtual machine, and those applications may link libraries or other code or the like, e.g., for a user interface, cryptography, communications, device drivers, mathematical or analytical functions and so forth. The stack may also interact with data. The stack may also or instead interact with users, and so users may be considered assets.

The threat management facility may include entity models **162**. The entity models may be used, for example, to determine the events that are generated by assets. For example, some operating systems may provide useful information for detecting or identifying events. For examples, operating systems may provide process and usage information that accessed through an API. As another example, it may be possible to instrument certain containers to monitor the activity of applications running on them. As another example, entity models for users may define roles, groups, permitted activities and other attributes.

The event collection facility **164** may be used to collect events from any of a wide variety of sensors that may provide relevant events from an asset, such as sensors on any of the compute instances **10-26**, the application protection facility **150**, a cloud computing instance **109** and so on. The events that may be collected may be determined by the entity models. There may be a variety of events collected. Events may include, for example, events generated by the enterprise facility **102** or the compute instances **10-26**, such as by monitoring streaming data through a gateway such as firewall **10** and wireless access point **11**, monitoring activity of compute instances, monitoring stored files/data on the compute instances **10-26** such as desktop computers, laptop computers, other mobile computing devices, and cloud computing instances **19, 109**. Events may range in granularity. An exemplary event may be communication of a specific packet over the network. Another exemplary event may be identification of an application that is communicating over a network.

The event logging facility **166** may be used to store events collected by the event collection facility **164**. The event logging facility **166** may store collected events so that they can be accessed and analyzed by the analytics facility **168**. Some events may be collected locally, and some events may be communicated to an event store in a central location or cloud facility. Events may be logged in any suitable format.

Events collected by the event logging facility **166** may be used by the analytics facility **168** to make inferences and observations about the events. These observations and inferences may be used as part of policies enforced by the security management facility. Observations or inferences about events may also be logged by the event logging facility **166**.

When a threat or other policy violation is detected by the security management facility **122**, the remedial action facility **128** may be used to remediate the threat. Remedial action may take a variety of forms, non-limiting examples including collecting additional data about the threat, terminating or modifying an ongoing process or interaction, sending a warning to a user or administrator, downloading a data file with commands, definitions, instructions, or the like to remediate the threat, requesting additional information from the requesting device, such as the application that initiated the activity of interest, executing a program or application to remediate against a threat or violation, increasing telemetry or recording interactions for subsequent evaluation, (continuing to) block requests to a particular network location or locations, scanning a requesting application or device, quarantine of a requesting application or the device, isolation of the requesting application or the device, deployment of a sandbox, blocking access to resources, e.g., a USB port, or other remedial actions. More generally, the remedial action facility **122** may take any steps or deploy any measures suitable for addressing a detection of a threat, potential threat, policy violation or other event, code or activity that might compromise security of a computing instance **10-26** or the enterprise facility **102**.

While the above description of the threat management system **101** describes various threats typically coming from a source outside the enterprise facility **102**, it should be understood that the disclosed embodiments contemplate that threats may occur to the enterprise facility **102** by the direct actions, either intentional or unintentional, of a user or employee associated with the enterprise facility **102**. Thus, reference to threats hereinabove may also refer to instances where a user or employee, either knowingly or unknowingly, performs data exfiltration from the enterprise facility **102** in a manner that the enterprise facility **102** wishes to prevent.

FIG. 2 illustrates a computer system according to an example embodiment. In general, the computer system **200** may include a computing device **210** connected to a network **202**, e.g., through an external device **204**. The computing device **210** may be or include any type of network endpoint or endpoints as described herein, e.g., with reference to FIG. 1 above. For example, the computing device **210** may include a desktop computer workstation. The computing device **210** may also or instead be any suitable device that has processes and communicates over a network **202**, including without limitation a laptop computer, a desktop computer, a personal digital assistant, a tablet, a mobile phone, a television, a set top box, a wearable computer (e.g., watch, jewelry, or clothing), a home device (e.g., a thermostat or a home appliance controller), just as some examples. The computing device **210** may also or instead include a server, or it may be disposed on a server.

The computing device **210** may provide a physical or virtual device as a platform for any of the entities described in the threat management environment above with reference to FIG. 1. For example, the computing device **210** may be a server, a client, a threat management facility, or any of the other facilities or computing devices described therein. In certain aspects, the computing device **210** may be implemented using hardware (e.g., in a desktop computer), soft-

ware (e.g., in a virtual machine or the like), or a combination of software and hardware (e.g., with programs executing on the desktop computer). The computing device **210** may be a standalone device, a device integrated into another entity or device, a platform distributed across multiple entities, or a virtualized device executing in a virtualization environment.

The network **202** may include the network **105** described above, which may be any data network(s) or internetwork(s) suitable for communicating data and control information among participants in the computer system **200**. This may include public networks such as the Internet, private networks, and telecommunications networks such as the Public Switched Telephone Network or cellular networks using third generation cellular technology (e.g., 3G or IMT-2000), fourth generation cellular technology (e.g., 4G, LTE, MT-Advanced, E-UTRA, etc.), fifth generation cellular technology (e.g., 5G), WiMax-Advanced (IEEE 802.16m) and/or other technologies, as well as any of a variety of corporate area, metropolitan area, campus or other local area networks or enterprise networks, along with any switches, routers, hubs, gateways, and the like that might be used to carry data among participants in the computer system **200**. The network **202** may also include a combination of data networks, and need not be limited to a strictly public or private network.

The external device **204** may be any computer or other remote resource that connects to the computing device **210** through the network **202**. This may include threat management resources such as any of those contemplated above, gateways or other network devices, remote servers or the like containing content requested by the computing device **210**, a network storage device or resource, a device hosting malicious content, or any other resource or device that might connect to the computing device **210** through the network **202**.

The computing device **210** may include a processor **212**, a memory **214**, a network interface **216**, a data store **218**, and one or more input/output devices **220**. The computing device **210** may further include or be in communication with peripherals **222** and other external input/output devices **224**.

The processor **212** may be any as described herein, and may generally be capable of processing instructions for execution within the computing device **210** or computer system **200**. The processor **212** may include a single-threaded processor or a multi-threaded processor. The processor **212** may be capable of processing instructions stored in the memory **214** or on the data store **218**.

The memory **214** may store information within the computing device **210** or computer system **200**. The memory **214** may include any volatile or non-volatile memory or other computer-readable medium, including without limitation a Random-Access Memory (RAM), a flash memory, a Read Only Memory (ROM), a Programmable Read-only Memory (PROM), an Erasable PROM (EPROM), registers, and so forth. The memory **214** may store program instructions, program data, executables, and other software and data useful for controlling operation of the computing device **200** and configuring the computing device **200** to perform functions for a user. The memory **214** may include a number of different stages and types for different aspects of operation of the computing device **210**. For example, a processor may include on-board memory and/or cache for faster access to certain data or instructions, and a separate, main memory or the like may be included to expand memory capacity as desired.

The memory **214** may, in general, include a non-volatile computer readable medium containing computer code that,

15

when executed by the computing device **200** creates an execution environment for a computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, or a combination of the foregoing, and/or code that performs some or all of the steps set forth in the various flow charts and other algorithmic descriptions set forth herein. While a single memory **214** is depicted, it will be understood that any number of memories may be usefully incorporated into the computing device **210**. For example, a first memory may provide non-volatile storage such as a disk drive for permanent or long-term storage of files and code even when the computing device **210** is powered down. A second memory such as a random-access memory may provide volatile (but higher speed) memory for storing instructions and data for executing processes. A third memory may be used to improve performance by providing even higher speed memory physically adjacent to the processor **212** for registers, caching and so forth.

The network interface **216** may include any hardware and/or software for connecting the computing device **210** in a communicating relationship with other resources through the network **202**. This may include remote resources accessible through the Internet, as well as local resources available using short range communications protocols using, e.g., physical connections (e.g., Ethernet), radio frequency communications (e.g., WiFi), optical communications, (e.g., fiber optics, infrared, or the like), ultrasonic communications, or any combination of these or communications through any other media that might be used to carry data between the computing device **210** and other devices. The network interface **216** may, for example, include a router, a modem, a network card, an infrared transceiver, a radio frequency (RF) transceiver, a near field communications interface, a radio-frequency identification (RFID) tag reader, or any other data reading or writing resource or the like.

More generally, the network interface **216** may include any combination of hardware and software suitable for coupling the components of the computing device **210** to other computing or communications resources. By way of example and not limitation, this may include electronics for a wired or wireless Ethernet connection operating according to the IEEE 802.11 standard (or any variation thereof), or any other short or long range wireless networking components or the like. This may include hardware for short range data communications such as Bluetooth or an infrared transceiver, which may be used to couple to other local devices, or to connect to a local area network or the like that is in turn coupled to a data network **202** such as the Internet. This may also or instead include hardware/software for a WiMax connection or a cellular network connection (using, e.g., CDMA, GSM, LTE, 5G, or any other suitable protocol or combination of protocols). The network interface **216** may be included as part of the input/output devices **220** or vice-versa.

The data store **218** may be any internal memory store providing a computer-readable medium such as a disk drive, an optical drive, a magnetic drive, a flash drive, or other device capable of providing mass storage for the computing device **210**. The data store **218** may store computer readable instructions, data structures, program modules, and other data for the computing device **210** or computer system **200** in a non-volatile form for subsequent retrieval and use. For example, the data store **218** may store the operating system, application programs, program data, databases, files, and other program modules or other software objects and the like.

16

The input/output interface **220** may support input from and output to other devices that might couple to the computing device **210**. This may, for example, include serial ports (e.g., RS-232 ports), universal serial bus (USB) ports, optical ports, Ethernet ports, telephone ports, audio jacks, component audio/video inputs, HDMI ports, and so forth, any of which might be used to form wired connections to other local devices. This may also or instead include an infrared interface, RF interface, magnetic card reader, or other input/output system for coupling in a communicating relationship with other local devices. It will be understood that, while the network interface **216** for network communications is described separately from the input/output interface **220** for local device communications, these two interfaces may be the same, or may share functionality, such as where a USB port is used to attach to a WiFi accessory, or where an Ethernet connection is used to couple to a local network attached storage.

The computer system **200** may include a peripheral **222** for the computing device **210** such as any device used to provide information to or receive information from the computing device **200**. This may include human input/output (I/O) devices such as a keyboard, a mouse, a mouse pad, a track ball, a joystick, a microphone, a foot pedal, a camera, a touch screen, a scanner, or other device that might be employed by the user **230** to provide input to the computing device **210**. This may also or instead include a display, a speaker, a printer, a projector, a headset or any other audiovisual device for presenting information to a user. The peripheral **222** may also or instead include a digital signal processing device, an actuator, or other device to support control of or communications with other devices or components. Other I/O devices suitable for use as a peripheral **222** include haptic devices, three-dimensional rendering systems, augmented-reality displays, magnetic card readers, and so forth. In one aspect, the peripheral **222** may serve as the network interface **216**, such as with a USB device configured to provide communications via short range (e.g., Bluetooth, WiFi, Infrared, RF, or the like) or long range (e.g., cellular data or WiMax) communications protocols. In another aspect, the peripheral **222** may provide a device to augment operation of the computing device **210**, such as a global positioning system (GPS) device, a security dongle, or the like. In another aspect, the peripheral may be a storage device such as a flash card, USB drive, or other solid-state device, or an optical drive, a magnetic drive, a disk drive, or other device or combination of devices suitable for bulk storage. More generally, any device or combination of devices suitable for use with the computing device **200** may be used as a peripheral **222** as contemplated herein.

Other hardware **226** may be incorporated into the computing device **200** such as a co-processor, a digital signal processing system, a math co-processor, a graphics engine, a video driver, and so forth. The other hardware **226** may also or instead include expanded input/output ports, extra memory, additional drives (e.g., a DVD drive or other accessory), and so forth.

A bus **232** or combination of busses may serve as an electromechanical platform for interconnecting components of the computing device **200** such as the processor **212**, memory **214**, network interface **216**, other hardware **226**, data store **218**, and input/output interface. As shown in the figure, each of the components of the computing device **210** may be interconnected using a system bus **232** or other communication mechanism for communicating information.

Methods and systems described herein can be realized using the processor **212** of the computer system **200** to

execute one or more sequences of instructions contained in the memory 214 to perform predetermined tasks. In embodiments, the computing device 200 may be deployed as a number of parallel processors synchronized to execute code together for improved performance, or the computing device 200 may be realized in a virtualized environment where software on a hypervisor or other virtualization management facility emulates components of the computing device 200 as appropriate to reproduce some or all of the functions of a hardware instantiation of the computing device 200.

FIG. 3 depicts a system 300 for monitoring user policies and security settings for electronic device usage based on a user status according to an example embodiment. The system 300 includes a threat management facility 301 which may include some or all of the features of the threat management facility 100 described previously. As shown, the threat management facility 301 includes several of the sub-facilities shown in FIG. 1 and described previously, such as a security management facility 322 (similar or the same as the security management facility 122), a policy management facility 312 (similar or the same as the policy management facility 112), a network access facility 324 (similar or the same as the network access facility 124), an entity model facility 362 (similar or the same as the entity model facility 162), an event collection and/or logging facility 364 (similar or the same as the combination of the event collection facility 164 and the event logging facility 166), and an action facility 328 (similar of the same as the remedial action facility 128). Each of the sub-facilities 322, 312, 324, 362, 364, 328 may include one or more data repositories 350 for retaining data associated with a respective facility.

As shown, the threat management facility 301 includes an employment status module 370 which is shown as a system which is in operable communication with the various sub-facilities 322, 312, 324, 362, 364, 328. In particular, the employment status module 370 may collect data from the various data repositories 350 of the sub-facilities 322, 312, 324, 362, 364, 328 for usage in accordance to the principles described herein. Likewise, the employment status module 370 may leverage the functionality of the various sub-modules 322, 312, 324, 362, 364, 328 in order to perform responsive actions in light of changes to employment status of users associated with the threat management facility 301 (users of an enterprise facility, for example).

The threat management facility 301 may further receive data from, and send data to, various other systems such as enterprise facility 302, including a managed device 322 having a monitoring agent S, and an enterprise administrator 321. It should be understood that while only one managed device 322 and enterprise administrator 321 is shown, the enterprise facility can comprise any number of endpoints, devices administrators and the like, such as the endpoints 10-26 shown in FIG. 1. Moreover, the threat management facility 301 may send data to, and receive data from, marketplace provider 399, such as a human resources software application provider or service.

The employment status module 370 may include one or more computer systems having one or more computer processors configured to execute instructions for performing a methodology as described herein. In particular, the employment status module 370 may be configured to associate respective security settings for a plurality of users of electronic devices, such as the monitored device 322, with employment statuses of the users. The employment status module 370 may receive information from one or more sources indicating a change in employment status for a user,

and use this received information to determine a change in security settings, automatically and/or responsive to receiving the information. The employment status module 370 may further be configured to apply the change in security settings, and then use the change to restrict electronic device usage of the user based on the change to the security settings of the user.

The employment status module 370 may further be configured to deploy and interact with a monitoring agent, such as the monitoring agent S, on a device operated by a user. The monitoring agent may be configured to monitor and/or control device usage by the user, with the direction of the employment status module 370 and instructions therefrom. The employment status module 370 may be configured to control the monitoring agent based on the employment status of the user and associated security settings as determined by the employment status.

To accomplish the above-described functionality, the employment status module 370 may communicate with the security management facility 322, which may be a sub-facility of the threat management facility 301. The security management facility 322 may incorporate services related to reviewing and analyzing and processing the information received from the monitoring or security agent S. The security management facility 322 may further be utilized in interacting with other endpoints (not shown) of the enterprise facility 302, such as firewalls or the like, as part of the restriction or actions taken in response to a change in security settings or monitoring status resulting from an employment status change for a user.

The employment status module 370 may further communicate with the policy management facility 312. The policy management facility 312 may manage rules or policies, permissions and the like, associated with individual devices and users. Blacklist (e.g., a block list) and whitelist (e.g., an allowed list) policies related to application permissions and the like may be associated with user policies and generated, stored and updated by the policy management facility 312. Resources from the policy management facility 312 may be leveraged in order to generate the key rules or associations underlying the employment status module 370 related to the association of security settings with employment status for users. Thus, the employment status module 370 may provide the resources allowing the employment status module 370 to dynamically and immediately respond in real time to changes in employment status with corresponding changes to security settings. The policy management facility 312 may originate and/or push information to the monitored device 322, the enterprise administrator 321 or any other device or endpoint of the enterprise facility 302.

The employment status module 370 may communicate with the network access facility 324. The network access facility 324 may be used to keep track of blacklists (e.g., the block lists), whitelists (e.g., the allowed lists), for network access requests made by user devices. These network access requests may be separate from the application permissions accounted for by the policy management facility 312. In other embodiments, these different categories of blacklist and whitelist information or data may be combined into a single repository or database related to devices usage, network access, and the like.

The employment status module 370 may communicate with the entity model facility 362. The entity model facility 362 may contain group-related information associated with particular entities such as the enterprise facility 302. The threat management facility 301 may be configured to manage threats and user status in the manner described herein on

more than one entity (i.e. enterprise facility) simultaneously. The entity model facility **362** may contain entity-specific rules which may be used by the employment status module **370** in applying rules or algorithms in determining security settings based on employment status.

The employment status module **370** may communicate with the event collection and logging facility **364**. As described with respect to FIGS. **6** and **8**, the event logging facility **364** may include resources for receiving events, event streams, event vectors and the like, and processing and storing this information in the employment status module **370**, and in the various sub-facility data repositories **350** as needed so that the employment status module **370** may be capable of using received information in updating security settings and monitoring status based on employment status.

The employment status module **370** may communicate with the action facility **328**. The action facility **328** may facilitate the interactions of the employment status module **370** with the various monitored devices (i.e. the monitoring or security agents associated therewith), enterprise administrators of an enterprise facility associated with the monitored devices and monitored users, or other endpoints of the enterprise facility. For example, other endpoints may include routers, switches, firewalls or other devices or virtual devices which are included in the enterprise facility and can be leveraged to enable the security settings determined by the employment status module **370**. For example, it is possible that security settings associated with a particular user could require instructions or settings to be changed in a firewall of the enterprise facility. Whatever the action, the action facility **328** may be utilized to accomplish the policy changes described herein.

FIG. **4** depicts an employment status module **470**, which may comprise one or more computer systems in accordance with an example embodiment. Embodiments of the computer system may include a module structure **430** that includes one or more modules or sub-modules, including an event collection and logging module **431**, a policy management module **432**, an agent management module **433**, an entity model module **434**, a device access module **435** and an action module **436**. A “module” may refer to a hardware based module, software based module or a module may be a combination of hardware and software. Embodiments of hardware based modules may include self-contained components such as chipsets, specialized circuitry and one or more memory devices, while a software-based module may be part of a program code or linked to the program code containing specific programmed instructions, which may be loaded in the memory device of the computer system. A module (whether hardware, software, or a combination thereof) may be designed to implement or execute one or more particular functions or routines. While the module structure **430** as a single entity, it should be understood with reference to FIG. **3**, that the resources embodied by the computer system of the employment status module **470** and module structure **430** thereof may be found across a larger facility architecture having many sub-facilities contributing resources and/or information to the module structure **430**.

Embodiments of the event collection and logging module **431** may include one or more components of hardware and/or software program code for receiving, processing and/or storing information indicating a change in an employment status of a user. This information may be received from various sources, including administrators of an enterprise facility or threat management facility, or from a marketplace provider, such as a third party human resources system or service associated with an enterprise facility. The employ-

ment status information may also be received directly from a monitored device (i.e. from a monitoring agent operable thereon).

The event collection and logging module **431** may further be configured for receiving, from a monitoring agent deployed on a monitored device, information related to device usage of the electronic device by the user. This type of device usage may relate to application usage, website usage, application installation or any other type of device usage which might be relevant in setting security settings. This information may further include information indicating the user has downloaded sensitive or confidential information from a repository associated with the at least one entity. Still further, the monitoring agent may be configured to provide information related to when the user attempts to perform data exfiltration with the electronic device, or access an application or website which allows for data exfiltration. Thus, the received information from the monitoring agent may relate to or include data flow information to and from the electronic device of a user.

The received information by the event collection and logging module **431** may be considered “events” as described herein, and may be received and processed by the event collection and logging module **431** according to the principles described in FIGS. **6** and **8**.

Embodiments of the policy management module **432** may include one or more components of hardware and/or software program code for determining a change to security settings for the user based on the change in the employment status of the user. Generally, the policy management module **432** of the employment status module **470** may include the analysis engine, logic, algorithms, ruleset, or model for providing the determinations and/or predictions made by the employment status module **470**. The policy management module **432** may further be configured to initiate the applying of changes to the security settings for the user in a security management system of the threat management facility based on these determinations.

The policy management module **432** may be configured to further escalate or increase a level of monitoring of a user based on received information from a monitoring agent of a monitored device of a user. This monitoring status or other type of user status of the user may not alone preclude the user from engaging in activity, but may be driven by a determination by the policy management module **432** based on received information that enhanced monitoring of a user or monitored device is appropriate.

Thus, the policy management module **432** may be configured to process updates in employment status of users and change the security settings for the user based on predetermined rules, logic, algorithms, logic or models.

In some embodiments, the policy management module **432** may deploy one or more machine learning models, described with reference to FIGS. **9** and **13**. Machine learning models of the policy management module **432** may be configured to predict a likelihood of one or more of the plurality of users to attempt to perform unauthorized data exfiltration and suggest a restriction for data processing activity for the one or more of the plurality of users to an administrator of the at least one entity based on the prediction. Machine learning models may further be configured to suggest or deploy a user policy, security settings, or the like, based on predictions output.

Embodiments of the agent management module **433** may include one or more components of hardware and/or software program code for deploying and controlling one or more monitoring agents on endpoints associated with a

client or enterprise facility (i.e. “monitored devices”). Thus, the agent management module **433** may be configured to deploy a monitoring agent on an electronic device of a user of the plurality of users such that the monitoring agent is capable of monitoring device usage of the electronic device and providing information back to the employment status module **470** related to the monitored device usage. The agent management module **433** may be configured to control the monitoring agent based on an employment status of the user and security settings for the user as determined using the employment status model **470** and the policy management module **432** thereof. The agent management module **433** may be configured to inform and/or apply changes to the monitoring agent that correspond to changes to the security settings for the user. Thus, the agent management module **433** may generally be configured to deploy, interact with, update and send and receive information with the monitoring agent of a monitored device.

Embodiments of the entity model module **434** may include one or more components of hardware and/or software program code for maintaining entity-specific data for application by the rules, logic, algorithms, logic or models of the policy management module **432**. Thus, the employment status module **470** may be configured to be deployed and operable for monitoring many entities simultaneously. The entity model module **434** may store and maintain the data associated with entity specific security settings, user classifications, and the like.

Embodiments of the device access module **435** may include one or more components of hardware and/or software program code for storing and maintaining information related to blacklists, whitelists and the like. Thus, the device access module **435** may be configured to maintain a list of applications and websites. This list or data may be utilized by the policy management module **432** in obtaining respective security settings for each of the plurality of employment statuses include settings for allowing, blocking or restricting the use of the applications and websites. Thus, the device access module **435** may include information associated with blacklisting the use of one or more applications or websites.

Embodiments of the action module **436** may include one or more components of hardware and/or software program code for performing actions by the employment status module **470** when appropriate, as determined by the policy management module **432**. For example, the action module **436** may be configured to restrict use of at least one data processing electronic device activity on an electronic device of the user based on the change to the security settings for the user. This restriction may be created by sending instructions or an update to the monitoring agent and/or any other devices within an enterprise facility (e.g. firewalls etc.) which would restrict device usage in the manner determined by the policy management module **432**.

In various embodiments, the action module **436** may specifically be configured to prevent data exfiltration based on the change to the security settings for the user. For example, the action module **436** may provide instructions to a monitoring agent which may blacklist the use of one or more known data exfiltration applications and/or websites, and/or the downloading of such known data exfiltration applications or websites. The action module **436** may be configured to alert an administrator of the at least one entity that the user has attempted to perform data exfiltration with the electronic device.

Thus, the action module **436** may be configured to originate actions which may be carried out by agent management module **433** associated with restricting data processing elec-

tronic device activity on the electronic device of the user based on the change to the security settings for the user. The action module **436** may further be configured to administer and carry out actions which, when implemented, enhance or increase the level of the monitoring of device usage of a monitored device. Likewise, the action module **436** may be configured to facilitate the performance and implementation of any action described herein attributable to the employment status module.

The employment status module **370** and the action module **363** may be configured to perform actions automatically, immediately and and responsively to the receiving the information indicating the change in the employment status of the user. Thus, the employment status module **370** may be configured to automatically adapt policy and security setting updates immediately and responsively as new employment status information is received.

Referring still to FIG. 4, embodiments of the computer system embodied by the employment status module **470** may be equipped with one or more memory devices **442** which may store the location information, information related to the above described tasks of the employment status module **470**, and a processor **441** for implementing the tasks associated with the employment status module **470**. The employment status module **470** further includes a data repository **450**. The data repository **450** may be a data collection area within a threat management facility. The data repository **450** may be a dedicated data repository associated with the employment status module **470**, but may also be a combination of one or more data repositories of the various sub-facilities of the threat management facility within which the employment status module **470** is employed (e.g. the data repositories **350**), or a combination thereof.

FIG. 5 depicts a managed device environment **500** in accordance with an example embodiment. As shown, the managed device environment includes a managed device **520**, such as an end point, compute instance, user device or the like running an operating system **570**. The managed device **520** is connected, for example, over a network, to a threat management facility **510** (which may be the same or similar to the threat management facilities **100**, **301** described hereinabove), an enterprise facility administrator associated with an enterprise facility, and an external data exfiltration application or website **512**. The managed device **520** may be a device that is part of the enterprise facility (not shown) overseen by the enterprise facility administrator, or may be an external mobile device, endpoint or compute instance (such as the mobile device **26**).

The managed device **520** may include a monitoring agent **530** which includes various features including an event detecting feature **540** that produces data, which are recognized as events by the monitoring agent **530**, the threat management facility **510** and the employment status module. The event detecting feature **540** of the monitoring agent **530** may be thus a source of event information.

The monitoring agent **530** may be an application or software which operates at an application layer on the managed device **520** and may be in operable communication with other applications **531a**, **531b**, **531c** or processes running on the managed device **520**. The monitoring agent **530** may be an application which is deployed by the threat management facility **510**, and may be in continual or regular communication with the threat management facility **510**.

The information gathered by the monitoring agent **530** may relate to device usage of the managed device **520**, particularly as the device usage relates to the other applications **531a**, **531b**, **531c** or processes running on the managed

device. While not shown, the monitoring agent **530**, and the event detecting feature **540** thereof, may be embodied by a software process, but may also include hardware (sensors) or other devices, or be embodied by a combination of hardware and software. The device usage may particularly pertain to instances when a user uses the device in a manner relating to data exfiltration and/or data downloading. For example, device usage may relate to a user downloading or otherwise using one of the applications **531a**, **531b**, **531c** or the external data exfiltration application or website **512**. Device usage may further pertain to instances when the user attempts to use the device to download sensitive information from the enterprise facility associated with the managed device **520**.

The output of the monitoring agent **530** may be recognized as events **544**. There may be many events provided by the monitoring agent **530**. The events may be processed by a local event processing facility **554** of the monitoring agent **530**. The event processing may perform processing and tokenization, if necessary. Some events may be recognized and evaluated in real-time, other events may be evaluated in the context of other events. Processing of the events include, without limitation, stream processing and/or bulk processing. Events may have attributes, or associated contextual information (e.g., mandatory, optional (e.g., best effort), sensitive (e.g., tokenize it in local event store). One or more of the events may be sent in real time and may be stored locally. Locally stored events may be retrievable by the local event processing facility **554** for further processing and/or communication. The events may be determined to be of interest based on policies put in place related to employment status of a user account associated with the managed device **520**.

A local event recorder **550** may be part of the local event processing facility **554**. Some recorded events may be stored locally, whereas some may be communicated to another compute instance, such as the threat management facility **510** and/or the employment status module. An event filter **562** may be used to parse the events locally. Local analytics **564** on the managed device **520** may be used to locally identify events of interest. A communication facility **560** of the managed device **520** and/or monitoring agent **530** communicate events to a threat management facility **510**, which may be a cloud facility.

Local enforcement **566** may be used to take steps in response to events, as determined by the policy management facility of the threat management facility **510**. The threat management facility **510** may communicate changes or updates to the local enforcement **566** based on determinations and changes of security settings and/or user policies made using the employment status module at the threat management facility **510**, as described above. Alternatively, the local enforcement **566** may be updated based on local policies or rulesets that are incorporated into the local monitoring agent **530** and processed locally at the local event processing facility **554**. This local processing by the monitoring agent **530** may allow the monitoring agent **530** to perform some or all of the functions of the threat management facility **510** and the associated employment status module locally without external connection or interaction with the threat management facility.

The local enforcement **566** may facilitate in the blacklisting of applications and/or websites or otherwise restrict usage of the monitored device **520**. The local enforcement **566** may be configured to implement or execute the steps necessary to prevent data exfiltration using the applications **531a**, **531b**, **531c** and/or the external data exfiltration appli-

cation or website **512**. Thus, the local enforcement **566** facilitate in preventing entire applications and/or websites or simply in the preventing certain functionality associated with those applications and/or websites, depending on the security settings and/or user policies.

FIG. 6 depicts an event collection environment **600** in accordance with an example embodiment. The event collection environment **600** may be used to receive and store events from various managed devices, endpoints or compute instances. Events are received at a threat management facility **610** by event collection **662**. Events may be received from managed devices, shown for the sake of clarity of illustration as a managed device **611** having a security or monitoring agent S, an enterprise facility administrator **612**, and a managed device **613** having a security or monitoring agent S, and/or a marketplace provider **680**, although events may be received from any number or type of managed devices. Events may be stored in the event store **664**, and also may be processed in real-time by the processing facility **666**. The employment status module **670** may be used to make observations and inferences based on the events, and support the automatic and responsive policy changes to security settings associated with user accounts based on the received events. The employment status module **670** may operate in conjunction with various facilities of the threat management facility **610**, including security management **671**, policy management **672** and network access **673** in order to provide for the functionality of restricting device activity based on the employment status of a user.

Received events from managed devices **611**, **613** may relate to device usage, applications operating on the operating system of the managed devices, websites visited by the managed devices, policy changes made locally by the monitoring agent, or any other monitoring or security information sent by the managed devices **611**, **613** to the threat management facility **610**. Received events from the enterprise facility administrator may relate to policy or security settings associated with user or device accounts made at the administrator level of the enterprise facility. Received events from the marketplace provider, which may be, for example, a third party human resources software or service, may relate to changes to employment status associated with a given user account, which may be received by the threat management facility for automatic processing by the employment status module **670**.

FIG. 7 depicts a policy management table **700** in accordance with an example embodiment. The policy management table **700** may be representative of a table used by the employment status module, as described herein, containing rules and status information associated with various users and user devices. The policy management table **700** may incorporate the data used by the employment status module, as described herein. While the policy management table **700** is shown as a single table, it should be understood that the employment status module may architecturally utilize any number of tables with any number of rows and columns for deploying the rules and algorithms and models described herein. Moreover, while the particular columns shown illustrate one embodiment of the types of data relied upon by the employment status module, various other types of data associated with categories of information are contemplated.

In some embodiments, the policy management table **700** may be a table compiled using, at least in part, data taken from data repositories associated with various different sub-facilities within a threat management facility. For example, the employment status module may be configured to use data from a policy management facility, such as the

25

policy management facility **112, 312**, including data associated with permissions and policies. Similarly, the employment status module may be configured to use data from a network access facility, such as the network access facility **124, 324**, such as block lists, white lists, black lists, and the like. The employment status module may be similarly configured to use data from entity models, such as the entity model facility **162, 362**, associated with entity models defining groups, roles, permitted activities and other attributes related to entities and/or users thereof.

As shown, the policy management table **700** includes six columns: a user column **702**, a device(s) column **704**, an entity column **706**, an employment status column **708**, a security settings column **710** and a user setting **712** column. The policy management table **700** includes six different primary rows associated with six different users 1-6.

The first column **702** relates to the user column. The users 1-6 may represent user accounts, which may be stored by the threat management facility pursuant to various other security offerings provided by the threat management facility, and retrieved therefrom by the employment status module for the population of the first column **702**. Alternatively, the first column **702** may be generated by the employment status module in accordance with a specific data entry during the setting up of the employment status module for a particular enterprise facility.

Each user shown in column **702** may include one or more devices associated with the user account, shown in the second column **704**. For example, as shown, users 1, 3 and 6 each include two separate devices. The devices listed in the second column **704** may each be devices associated with each user which a monitoring agent, as described above, is deployed thereon.

The users provided in the first column **702** may relate to different enterprise facilities, customers, or clients, in some embodiments. As shown in the third column **706**, the entity model may store information associated with which entity (i.e. enterprise facility, customer or client), a user is associated with. In the embodiment shown, the policy management table **700** provides six different users spanning across two different entities.

The fourth column **708** is shown, which keeps track of employment statuses for each of the user accounts. This employment status information may be input directly by an administrator of the threat management facility or the enterprise facility, or may be automatically acquired or scraped from a human resources marketplace provider third party platform, or the like. Employment statuses shown include employed, departing, or terminated, but these are non-limiting examples. The various employment statuses associated with each user profile are particularly used by the employment status module in order to determine security settings and/or user settings, as shown in columns **710, 712**.

The fifth column **710** shown are codes associated with particular security settings associated with each user profile. These security settings relate to the employment status, and may be generated pursuant to an algorithm, ruleset, or model as provided by the employment status module. In some cases, the security settings in the fifth column **710** are associated with a given user account, such as an E1 ruleset (i.e. employed, first setting) of the first user. The denotation E1 may be representative of any denotation calling back to one or more security settings stored in the employment status module associated with a particular employment status. In this case, the E1 ruleset may be applied across the entire employee account, regardless of the user device. In other cases, the security settings may be more particularly

26

tailored by device. For example, user 3, who is departing, has a D1 ruleset (departing, first setting) when using a first device, but may include a D2 ruleset (departing, second setting) when using a second device. A ruleset may correspond to both an employment status, (i.e. the letters for each of the security settings correspond to the employment status), but there may be variations for each setting depending on a plurality of other features associated with a given user account. For example, if a departing user at a management level may have a different ruleset relative to a departing user at a staff level. Thus, while a change in employment status will generally trigger a change in a ruleset by the employment status module, different employees at different levels and in different situations may be treated differently, even sharing the same employment status.

Finally, an additional user setting is shown in a sixth column **712**. The sixth column **712** includes, for example, a monitoring status associated with a degree of monitoring. This user setting may be either related or unrelated to an employment status, and may escalate based on received events and user activity. While the policy management table **700** may include three different monitoring settings, of low, medium, and high, these are exemplary.

While the policy management table **700** shown provides an example of a data table which may be compiled and used by algorithms, rules or models of the employment status module, any amount of data may be used by the employment status module. Whatever data utilized, security settings and security policy may be considered to automatically change across each row (user profile) when a change is received or made to the employment status column (i.e. the fourth column **708** shown in the policy management table **700**).

FIG. 8 depicts a system **800** for monitoring user status and response in accordance with an example embodiment. In general, the system may include one or more monitored devices **802** that use local monitoring agents **808** to gather events **806**, which may be expressed data, vector data, or the like. The monitored devices **802** and/or monitoring agents **808** thereof may then report these events **806** to a threat management facility **812**. The threat management facility **806** may store the events **806** as a data stream **814** in a data repository **816** such as one or more memory spaces or other data stores of the threat management facility **812**. The event stream **814** may be analyzed with the employment status module **820** and/or the interactions with the various sub-facilities of the threat management facility **812**.

The monitored devices **802** may be any of the types of monitored devices and/or user devices or compute instances described herein, including without limitation any physical device such as a laptop, desktop, gateway, router, firewall, smartphone, tablet, or the like, as well as a virtualized instance of any of the foregoing or any other computer, user device, container or the like. The events **806** may also generally be any of the events described herein. The monitoring agent **808** may be any of the security or monitoring agents described herein, or any other software component or the like executing on or in association with one of the monitored devices **802** to locally manage monitoring and security of monitored device and/or coordinate security and monitoring services with the threat management facility **812** and other remote resources.

The local monitoring agent **808** may collect events **806** from sensors or any combination of software and hardware systems operable on the monitored device **802**, and form the collected events **806** into event vectors, where applicable, for communication to the threat management facility **812**. The local monitoring agent **808** may process events **806** in

a variety of ways in order to facilitate communication, computational efficiency, or downstream processing. For example, the local monitoring agent **808** may tokenize the collected events **806**. That is, a process that causes or creates an event **1406** may be assigned a number or other identifier, which may be used locally by a compute instance or globally within the enterprise to identify a particular, known process. An event **806** may also encode (tokenized or otherwise) a relationship among different processes. For example, for a particular process that caused an event **806**, a parent-child relationship or other dependency with other processes may be encoded by providing process identifiers or the like within the event **806**, along with information characterizing the relationship among the processes. A Uniform Resource Locator or other information for identifying resources or network locations may also be tokenized or otherwise processed to support efficiency, consistency and the like. For example, a URL may be encoded in an event **806** as a hash of a URL, or as a portion of a URL, or some combination of these (e.g., a literal encoding of the top level domain, and a hash of some or all of the remaining path information). Other events **806** such as registry changes, system calls, remote procedure calls and the like may be literally encoded into an event **806** where they are relatively compact, or identified using any suitable tokenization, compression, or the like.

Other techniques may also or instead be used. For example, user-specific or machine-specific information may be altered where appropriate to anonymize the events **806** and mitigate exposure of sensitive information during network communications. An event vector or individual events **806** therein, may also or instead be encrypted in order to secure the contents against malicious interception. In another aspect, the events **806** may be compressed to conserve network resources. The events **806** may also or instead be prioritized, e.g., in order to increase sensitivity and decrease response times for events **806** associated with a high likelihood of unwanted activity or data exfiltration. In this latter aspect, the local monitoring agent **808** may locally analyze events **806** in order to permit suitable prioritization, as well as to support local detection and response to malicious, or potentially malicious activity.

It will also be appreciated that events **806** may usefully be labelled in a variety of ways. While labeling with process identifiers is described above, this may also or instead include an identification of an entity associated with the event **806**. In this context, the entity may be any physical, logical, or conceptual entity useful for monitoring activity of the monitored devices **802** as described herein. For example, the entity may include a user, a physical device, a virtualized machine, an operating system, an application, a process, a hardware subsystem (e.g., a network interface card, USB drive, camera, etc.), a network resource, a domain controller, a remote software service, and so forth. It should also be understood that the various entity types may be concurrently associated with a particular event **806**, or particular events **806** may be associated with multiple entities or event vectors. Thus for example, storing or downloading a file may be an event **806** associated with a particular user, a particular device or machine, a particular operating system, a particular physical storage device, and so forth. Similarly, attempting to perform data exfiltration may, access a data exfiltration application or website, or attempting to download a data exfiltration application or website may be an event **806** associated with a particular user, a particular device or machine, a particular operating system, a particular physical storage device, and so forth.

In general, the source of each event **806**, such as a particular sensor, or some entity, computing object or the like associated with the event **806**, may be encoded with the event **806** to permit explicit identification by the threat management facility **812** or other downstream processing resources. Although depicted as having similar size, it will also be understood that the events vector **806** may contain data or comprise event vectors of any size, and may usefully encode any number of different events **806**.

The events **806** may be received by the threat management facility **812** and stored as an event stream **814** in a data repository **816**, which may be any data store, memory, file or the like suitable for storing the events **806** or event vectors. The events **806** may be time stamped or otherwise labeled by the threat management facility **812** to record chronology. The event stream **814** may be used for analysis and detection as further described herein. In general, an employment status module **820** may analyze the event stream **814**, in accordance with the methods, architecture, process steps and embodiments described herein, and produce actions **850** deployable on the monitored devices **802** and other facilities or devices of the threat management facility **812** in order to prevent data exfiltration based on a user employment status, as described herein.

While not shown, events received from devices other than monitored devices are also contemplated, which can be stored at the data repository **816** and processed as part of the event stream **814** at the threat management facility **812**. For example, a change in employment status event may be provided in the event stream **814** by an administrator of an enterprise facility, an administrator of a threat management facility, or a marketplace provider.

FIG. 9 depicts a machine learning environment **900** in accordance with an example embodiment. The machine learning environment **900** includes training data **902** provided for training **904** of a machine learning trained model **906** for outputting a prediction **910**. As contemplated herein, the machine learning may be deployed to predict a likelihood of a user to attempt to perform unauthorized data exfiltration, based on the machine learning trained model **906**. The machine learning trained model **906** may also further predict a likely response to a received event, for example. Still further, the machine learning trained model **906** may suggest or deploy a user policy, security settings, or the like, based on the prediction **910** output by the machine learning trained model **906**.

The training data **902** may be a data set of example instances of employment statuses, security settings associated with employment statuses, user statuses, received information from monitoring agents related to historical device usage, and any historical responses to that received information include actions taken, monitoring escalations or the like. The training data may be data unrelated to the enterprise facility, customer or client, and may be initial data from which the threat management facility, employment status module and/or machine learning model thereof, may learn from. In other embodiments, the training data may be accumulated from real data over time as the systems run pursuant to the above described approaches.

As shown, once the machine learning trained model **906** is trained, input data **908** may be provided to the machine learning trained model **906**, which may output predictions based on the learned model. As described, the prediction **910** may relate to a likelihood of a user to attempt to perform unauthorized data exfiltration, a likely response to a received event, or a likely desired security setting or desired degree of monitoring based on learned historical preferences.

The machine learning trained model **906** may, for example, be vector representations or the like of different events expected for or associated with a user or enterprise facility, and may also include information about the frequency, magnitude, or pattern of occurrence for each such event. As events are collected, a statistical model or the like may be developed for each event represented within the machine learning trained model **906** so that a baseline of expected activity can be created. In one aspect, an existing model may be used for the machine learning trained model **906**, e.g., when the user or enterprise facility is already known and well characterized. The entity model may also or instead be created by observing activity by a user or enterprise facility over time. This may include, for example, monitoring the entity for an hour, for a day, for a week, or over any other time interval suitable for creating a model with a sufficient likelihood of representing ordinary behavior to be useful as a baseline as contemplated herein. It will also be understood that, once the machine learning trained model **906** is created, it may usefully be updated, which may occur at any suitable intervals according to, e.g., the length of time to obtain a stable baseline, the amount of activity, or any other factors.

Once the machine learning trained model **906** has been created and a stable baseline established, the machine learning trained model **906** may be deployed for use in monitoring prospective activity of users and monitored devices and suggesting appropriate activity based on received events. This monitoring may, for example, use the same event stream **814** described hereinabove. It will be appreciated that the machine learning trained model **906** may generally be deployed as fixed or relatively static or discrete models, or any one or more of the machine learning trained models **906** may be continuously updated so that they change over time as new information becomes available, e.g., in the event stream **814** or otherwise.

The employment status modules described herein may compare new events **806** generated by a monitored device, as recorded in the event stream **814**, to the machine learning trained model **906** that characterizes a baseline of expected activity. This comparison may usefully employ a variety of vector or similarity measures known in the art. For example, the comparison may use one or more vector distances such as a Euclidean distance, a Mahalanobis distance, a Minkowski distance, or any other suitable measurement of difference within the corresponding vector space. In another aspect, a k-nearest neighbor classifier may be used to calculate a distance between a point of interest and a training data set, or more generally to determine whether an event should be classified as within the baseline activity characterized by the entity model.

FIG. **10** depicts a method **1000**, in accordance with an example embodiment, for monitoring user policies and security settings for electronic device usage based on a user status. The method **1000** may be performable by, for example, the one or more computer systems and/or processors of the threat management facility **100**, **301** described hereinabove. This method **1000** may be particularly advantageous in precluding the need for individualizing policy changes when a user is known to have a changed employment status. In particular, the method **1000** includes incorporating an employment status module which automatically, without human intervention, updates policy changes and security settings for a user which can be pushed to a user device and thereby protect or prevent user activity on monitored devices.

As shown in step **1002**, the method **1000** may begin with storing an employment status module, such as the employment status module **500** described hereinabove. The employment status module may be provided with a service-oriented architecture, or a microservice container-based architecture. The employment status module may incorporate sub-modules which span across multiple facilities of the threat management facility, such as a security management facility, an update management facility, a policy management facility, a network access facility, an entity model facility, an event collection facility, an event logging facility, and a remedial action facility.

The employment status module may be stored in association with at least one entity, such as a customer, client or enterprise facility. In other embodiments, separate employment status modules may be deployed and stored by the threat management facility for each different customer, client or enterprise facility in the form of a separate compute instance or virtualized instance.

At this step **1002**, the rules and/or algorithm associated with the employment status module may be input, provided, coded or otherwise created and stored at the threat management facility. The employment status module may associate a plurality of employment statuses with respective security settings for a plurality of users associated with the customer, client or enterprise facility. For example, employment statuses may include various types of statuses, such as employed, not employed, hired (but not yet employed), furloughed, under contract (contractor), subcontractor, pending termination, pending departure, or the like. Any type of employment status may be contemplated and provided for in the employment status module.

The various employment statuses may be assigned respective security settings within the employment status module. Security settings may particularly relate to the degree of monitoring, the ability to access certain websites and applications, the ability to access data sources of the customer, client or enterprise facility and/or permission status of a user. Each employment status may include a unique associated security setting. In some cases, the security settings may include statuses or settings associated with websites or applications, where the statuses or settings are, for example, whitelist status (allowed) and/or blacklist status (blocked or prohibited). The websites or applications may be websites or applications which provide data exfiltration opportunities if not blacklisted. In other cases, security settings may include statuses related to a degree of monitoring performed by a monitoring agent of the device and/or the threat management facility. In any event, the security settings may relate to data processing and/or data exfiltration activity associated with user accounts and/or the devices or compute instances, or the like, associated with those user accounts.

The method **1000** includes a second step **1004** of receiving information by the computer system and/or the threat management facility, information indicating a change in an employment status of user associated with the customer, client or enterprise facility. The received information may be received in accordance with the event collection environment, described hereinabove. The change in employment status may be provided by a change made by an administrator of the customer, client or enterprise facility to a user account or setting, which is then provided to the threat management facility and/or computer system as described herein in the form of an event. In other embodiments, the change in the employment status may be input by an administrator at the threat management facility.

31

In still other embodiments, the information may be received from a data scraping agent deployed at an enterprise facility which may be used to automatically determine employment statuses of users based on various other systems associated with the customer, client or enterprise facility. For example, a marketplace provider and/or cloud enterprise facility may be a human resources system provider which already contains data associated with employment status of users of the customer, client or enterprise facility, and may be provided to the threat management facility and/or computer systems thereof in accordance with the step 1004.

At a next step 1006, the threat management facility and/or computer system thereof determines whether an employment status change requires a change in security settings when new information is received. Step 1006 may be an optional step. In some instances, a change in employment status may not require a change in security settings such as a blacklisting of a website or application. In these instances, for example, a change in employment status may not be restrictive in nature (e.g. when an employment status changes from “contractor” to “employed”). If no change in security steps are warranted, at a step 1008 monitoring may continue, either as normal or in an escalated manner. For example, even if no prevention of usage is required by a change in a security setting, it may be possible to escalate monitoring in some manner by the threat management facility.

If it is determined that a change in security settings is required, a step 1010 may include determining a change to security settings for the user based on the change in the employment status of the user. For example, the step 1010 may include interfacing with the employment status module, and algorithms or rules associated therewith, in order to determine what changes to security settings are needed based on the received change in employment status.

At a step 1012, the method 1000 includes applying the determined change of security settings to a user account. Step 1012 may further include the interfacing with a network access facility, which may be a facility associated with actually performing the blocking to parse network access requests and apply the determined policies and security setting changes. Additionally, the step 1012 may include the computer system interfacing with a policy management facility to update a blacklist and/or whitelist associated with user accounts for particular websites and/or applications which may support data exfiltration or other unwanted user activity. At step 1012, the threat management facility and/or computer system thereof may communicate policy and security setting updates to a monitoring or security agent of devices associated with the user.

The step 1012 may further include a step 1014, of restricting the user from engaging in data processing activity on an electronic device associated with the user account and preventing data exfiltration based on the change to the security settings of the user. To accomplish step 1014, the computer system of the threat management facility may send a policy update to the security or monitoring agent on the user device. This policy update may update settings, which when interacted with by the network access facility, prevent the device from engaging in unwanted activity.

The processing at steps 1004-1014 may be performed automatically, immediately and/or responsively to the receiving the information indicating the change in employment status at step 1004. In some embodiments, the processing in steps 1004-1012 may be subject to a real-time constraint and may be conducted in real-time as responsive

32

awareness of the threat management facility and/or computer system thereof to the change in employment status.

FIG. 11 depicts a method 1100, in accordance with an example embodiment, of deploying a monitoring agent. Like the method 1000, the method 1100 may be performable by, for example, the one or more computer systems and/or processors of the threat management facility 100, 301 described hereinabove. The method 1100 may be particularly advantageous in monitoring a user device activity based on an employment status of the user, and in providing event information back to the threat management facility or an administrator of the client device for processing and policy decision making. The method 1100 may further advantageously provide for dynamic monitoring settings based on the employment status of the user. When combined with the method 1000, the method 1100 may further be configured to restrict device usage when a monitoring agent detects device usage which is not permitted by security settings associated with a user.

The method 1100 includes a first step 1102 of storing an employment status module, which may include the methodology and implementation described hereinabove with respect to the step 1002 of the method 1000. In the step 1102, the employment status module may particularly focus on algorithms, a ruleset and/or a trained model pertaining to escalating and de-escalating levels of monitoring provided by the monitoring agent based on employment statuses of users.

The method 1100 includes a second step 1104 of deploying a monitoring agent on an electronic device of a user. In some embodiments, the monitoring agent may be deployed on all devices or endpoints associated with a particular user. Likewise, the monitoring agent may be deployed on any and all devices or endpoints for any number of users associated with a customer, client or enterprise facility. The monitoring agent may be a security agent, application or the like required to run on the device of the user and in regular communication with the threat management facility.

The monitoring agent may be configured to monitor device usage of whatever device or endpoint the monitoring agent is deployed. This device usage information, data or stream of events may be provided in back to the threat management facility pursuant to the environment and architecture described hereinabove. The monitoring agent may also be configured to provide control and security features in order to enable any changes in security settings associated with a user based on the employment status of the user, as described in the method 1000.

The monitoring agent may be deployed with different levels of monitoring based on the status or security settings associated with a given user. For example, in one setting, device activity may not be monitored. In another setting, monitoring may be required. In another setting, device activity may be monitored only when particularly defined activity is being conducted on a monitored device. For example, if a data exfiltration application or website is being accessed or used by the device, the monitoring agent may be configured to provide information related to this event to an administrator of the enterprise facility or to the threat management facility for further processing. Whatever the monitored setting, the setting may be automatically be controlled and updated by the threat management facility responsive to when the employment status module determines that a change in employment status occurs with a given user.

The method 1100 includes a next step 1106 of determining security settings for the user based on a current employ-

33

ment status and then a step **1108** of applying the security settings by the monitoring agent. The step **1106** may be conducted at the threat management facility based on the current employment status of a user. Either before or after the monitoring agent is deployed in the step **1104**, the security settings for the individual user may be determined and applied by the threat management facility, thereby configuring the monitoring agent to monitor the device or endpoint upon which the monitoring agent is installed or operating.

At a step **1110**, the method **1100** includes receiving information indicating a change in an employment status of a user and/or determining a change in an employment status of a user based on received information. The step **1110** may be the same or similar to the steps **1004** and/or **1010** described previously. In some embodiments, the step **1110** may further include receiving changes to an employment status at the monitoring agent. However, in many cases, the employment status information may be received and processed at the threat management facility at the back end.

After receiving and/or determining a change in employment status, the method **1100** includes a final step **1112** of controlling the monitoring agent and/or device usage based on an employment status of the user and associated security settings for the user as determined during the step **1110**. As described previously, the monitoring agent may be controlled in various ways at step **1112** including, but not limited to, by changing a level of monitoring (e.g. increasing or decreasing the amount of information provided back to a system that the monitoring agent is connected to), or by changing a particular device usage control associated with the device being monitored (e.g. ensuring the device is prevented or allowed to access websites or applications, such as those which enable data exfiltration). Pursuant to the controlling, the monitoring agent may be in communication with the threat management facility, or an administrator of an enterprise facility, or the like.

FIG. **12** depicts a method **1200**, in accordance with an example embodiment, for preventing data exfiltration based on user status. The method **1200** provides one example of a method of escalating monitoring and/or restrictions based on received information and events. While the method **1200** provides one example of an escalation strategy, it should be understood that various strategies are contemplated that incorporate the principles described herein in any combination or order. The method **1200** may occur after a user has been configured up with a particular set of security settings based on employment status, as described above, and these settings have been provided to the monitoring agent. At start of the method **1200**, the device is being monitored, either as a baseline level of security, or because an employment status change has necessitated a certain degree of monitoring. However, the device has not yet been restricted.

The method **1200** includes a first step **1202** of receiving information related to device usage from a monitoring agent, for example, deployed on a device being used by a user of an enterprise facility. This receiving of information may occur at the threat management system, or at the monitoring agent locally running at the end point or device of the user. This received information may include an event such as a user downloading an application which provides for data exfiltration, or visiting a web page that provides for data exfiltration, or opening an account for a service which provides for data exfiltration. Events may also relate to downloading information from a system of the enterprise facility, customer or client. For example, if a user begins to download sensitive information from the enterprise facility

34

locally onto their device or endpoint, information related to such an event may be provided by a monitoring agent and received by the threat management facility. Whatever type of information is received, the method may proceed to responding automatically and responsively to this information, as described further below.

Based on this received information, the method **1200** includes a next step **1204** of escalating a user status based on the received information. The user status may relate to a status for a user profile associated the employment status module. For example, the user status may be stored or otherwise saved in a database at the threat management facility (e.g. a policy management facility). This user status may be independent of an employment status, but may relate to an employment status. For example, a user employment status may be "pending departure" (i.e. maybe working within a two week notice of leaving), but a monitoring status associated with this employment status may be escalated from "minimal" to "moderate." The employment status module may include security settings associated with each employment status and user status. Again, the change in the user status may occur automatically and responsively to the new information received related to the event which occurred whereby the information related to this event was received in step **1202**.

The method **1200** includes a further step **1206** of monitoring device usage pursuant to the escalation. This monitoring may particularly relate device usage which supports the uploading of data by an electronic device. In other embodiments, an escalation in device usage may relate to the downloading of data by the electronic device, the downloading, installing or using of an application or service, the visitation or interaction with a webpage, or the like.

During the monitoring, the method **1200** includes a next step **1208** of determining that the device usage is not appropriate. This determination may be made by deploying logic, algorithms, rules or models of the employment status module, as to the current security settings of the user based on the employment status and user settings. For example, the step **1208** may be a determination that the user is attempting to upload information to a data exfiltration application or website.

In a next step, the method **1200** includes a step **1210** of sending an alert to an administrator. The administrator may be an administrator of the enterprise facility, customer or client, or may also be an administrator of the threat management facility. Other types of alerts may be provided, for example, directly to the device of the user. Whatever the embodiment, the alerting step may be an optional step of the process, but may be an automated alert that arrives as a pop up window, an email communication, an alert within an administrator alert interface, or the like.

Finally, at steps **1212**, **1214** and **1216**, various measures may be taken to impact device usage based on the escalation. For example, the method may be used to restrict a user from data processing in particular manners **1212**, prevention of data exfiltration **1214** (or even data downloading), or specifically blacklist applications, websites or services **1216**. Various device changes may be deployed by the threat management facility, monitoring agent and combinations thereof.

FIG. **13** depicts a method **1300**, in accordance with an example embodiment, of predicting a likelihood of a user attempting to perform unauthorized data exfiltration. The method **1300** may be particularly advantageous in using machine learning in predicting problematic behavior and/or predicting likely responses or restrictions based on past or

35

historical data. The method **1300** may be implemented by one or more of the foregoing devices and/or systems mentioned previously and is discussed by way of reference thereto.

The method **1300** may include a first step **1302** of receiving training data and a second step **1304** of training a machine learning model with the training data. The training data received may be a data set of example instances of employment statuses, security settings associated with employment statuses, user statuses, received information from monitoring agents related to historical device usage, and any historical responses to that received information include actions taken, monitoring escalations or the like. The training data may be data unrelated to the enterprise facility, customer or client, and may be initial data from which the threat management facility, employment status module and/or machine learning model thereof, may learn from. In other embodiments, the training data may be accumulated from real data over time as the systems run pursuant to the above described approaches.

Whatever the embodiment, it is contemplated that machine learning may be deployed to predict a likelihood of a user to attempt to perform unauthorized data exfiltration, based on a machine learned model, at a step **1306**. The machine learning model may also further predict a likely response to a received event, for example. Finally, at a step **1308**, the machine learning model may suggest or deploy a user policy, security settings, or the like, based on the prediction.

The methods **1000-1300** are described as separate and distinct methods performable by a computer system in the manner described herein above. However, embodiments may combine various methods and/or steps within methods **1000-1300** into an aggregate methodology which incorporates the steps from one or more of the methods **1000-1300** described herein. For example, the method **1000** may incorporate each of the distinctive steps **1104**, **1112**, thereby including the deployment of the monitoring agent and the controlling of the monitoring agent. Similarly, the methodology in the method **1000** of applying the changes at step **1012** and restricting data processing in step **1014** may incorporate any and all of the specific methodology of the method **1200**. Likewise, the storing of the employment status module **1002** of the method **1000** may further include any of the machine learning methodology described in the method **1300**. It will be appreciated that various methodology combinations consistent with the present description are contemplated herein.

In this manner, the foregoing systems, devices, and methods provide a mechanism by which the possibility of data exfiltration is monitored and/or controlled based on the employment status of a user. As a user who has been recently terminated by his or her employer is likely to collect data generated by the employer, which may be sensitive and/or proprietary, the disclosed embodiments provide various means in which such data collection and/or exfiltration may be prevented. The described technologies may be particularly useful where the user may have obtained employment at a competitor of the employer's and the user is attempting to use data from the now-former employer at his or her future employer (i.e., the competitor). These technologies, and others described herein, increase the overall security within an organization's digital environment and reduces the likelihood of improper data exfiltration.

Although the foregoing Figures illustrate various embodiments of the disclosed systems and methods, additional and/or alternative embodiments are contemplated as falling

36

within the scope of this disclosure. For example, in one embodiment, this disclosure provides for a method comprising: storing, by a computer system at a threat management facility, an employment status module associated with at least one entity, where the employment status module associates a plurality of employment statuses with respective security settings for a plurality of users associated with the at least one entity, and the respective security settings relate to data processing electronic device activity by the plurality of users. The method may further include receiving, by one or more processors of the computer system, information indicating a change in an employment status of a user of the plurality of users and determining, by the one or more processors of the computer system using the employment status module, a change to security settings for the user based on the change in the employment status of the user. Still further, the method may include applying, by the one or more processors of the computer system, the change to the security settings for the user in a security management system of the threat management facility and restricting use of at least one data processing electronic device activity on an electronic device of the user based on the change to the security settings for the user.

In another embodiment of the method, a monitoring agent is located on the electronic device of the user, wherein the monitoring agent is configured to monitor device usage of the electronic device and provide information to the computer system related to the monitored device usage.

In a further embodiment of the method, the monitoring agent is configured to restrict the user from engaging in the at least one data processing electronic device activity on the electronic device of the user based on the change to the security settings for the user.

In yet another embodiment, the method includes controlling the monitoring agent, by the one or more processors of the computer system, based on the employment status of the user and the security settings for the user.

In yet a further embodiment, the method includes receiving from the monitoring agent, by the one or more processors of the computer system, information related to device usage of the electronic device by the user and escalating, by the one or more processors of the computer system, a user status of the user based on the received information from the monitoring agent.

In another embodiment of the method, the escalating the user status of the user based on the received information from the monitoring agent comprises: increasing the level of monitoring performed by the monitoring agent.

In a further embodiment of the method, the increasing the level of monitoring performed by the monitoring agent results in the monitoring agent monitoring of data flows to and from the electronic device of the user.

In yet another embodiment, the escalating the user status based on the received information from the monitoring agent comprises: changing the security settings for the user.

In yet a further embodiment, the information related to device usage of the electronic device by the user includes information indicating the user has downloaded sensitive or confidential information from a repository associated with the at least one entity.

In another embodiment, the monitoring agent is configured to monitor device usage which supports the uploading of data by the electronic device, and the restricting use of at least one data processing electronic device activity on an electronic device of the user based on the change to the security settings for the user comprises: preventing, by the

one or more processors of the computer system, data exfiltration based on the change to the security settings for the user.

In a further embodiment of the method, preventing data exfiltration based on the change to the security settings for the user comprises: blacklisting, by the monitoring agent, the use of one or more applications or websites.

In yet another embodiment of the method, the restricting the user from engaging in the at least one data processing electronic device activity on an electronic device of the user based on the change to the security settings for the user comprises: informing, by the one or more processors of the computer system, the monitoring agent of the change to the security settings for the user.

In yet a further embodiment, the monitoring agent is configured to provide information to the one or more processors of the computer system related to when the user attempts to perform data exfiltration with the electronic device.

In another embodiment, the method includes alerting, by the one or more processors of the computer system, an administrator of the at least one entity that the user has attempted to perform data exfiltration with the electronic device.

In a further embodiment, the device usage includes application usage, website usage and application installation.

In yet another embodiment, the employment status module maintains a list of applications and websites, and wherein the respective security settings for each of the plurality of employment statuses include settings for allowing, blocking or restricting the use of the applications and websites.

In yet a further embodiment, the employment status module includes a machine learning model, the method further comprises: predicting, by the machine learning model, a likelihood of one or more of the plurality of users to attempt to perform unauthorized data exfiltration; and based on the predicting, suggesting, by the one or more processors of the computer system, a restriction for data processing activity for the one or more of the plurality of users to an administrator of the at least one entity.

In another embodiment, the determining the change to security settings for the user based on the change in the employment status of the user, the applying the change to the security settings for the user in a security management system of the threat management facility, and the restricting the user from engaging in the at least one data processing electronic device activity on an electronic device of the user based on the change to the security settings for the user, are each performed automatically and responsively to the receiving the information indicating the change in the employment status of the user.

In another embodiment, the disclosure provides a computer system including one or more processors; one or more memory devices coupled to the one or more processors; and one or more computer readable storage devices coupled to the one or more processors, wherein the one or more storage devices contain program code executable by the one or more processors via the one or more memory devices to implement a method for changing security settings for a user based on employment status, the method comprising: storing, by the computer system at a threat management facility, an employment status module associated with at least one entity, wherein the employment status module associates a plurality of employment statuses with respective security settings for a plurality of users associated with the at least one entity, and the respective security settings relate to data

processing electronic device activity by the plurality of users. The method further includes receiving, by the one or more processors of the computer system, information indicating a change in an employment status of a user of the plurality of users, determining, by the one or more processors of the computer system using the employment status module, a change to security settings for the user based on the change in the employment status of the user. The method still further includes applying, by the one or more processors of the computer system, the change to the security settings for the user in a security management system of the threat management facility, and restricting use of at least one data processing electronic device activity on an electronic device of the user based on the change to the security settings for the user.

In another embodiment, the disclosure provides a computer program product for changing security settings for a user based on employment status, the computer program product comprising: one or more computer readable storage media having computer readable program code collectively stored on the one or more computer readable storage media, the computer readable program code being executed by one or more processors of a computer system to cause the computer system to perform a method comprising: storing, by the computer system at a threat management facility, an employment status module associated with at least one entity, wherein the employment status module associate a plurality of employment statuses with respective security settings for a plurality of users associated with the at least one entity, and the respective security settings relate to data processing electronic device activity by the plurality of users. The method further comprises receiving, by the one or more processors of the computer system, information indicating a change in an employment status of a user of the plurality of users, and determining, by the one or more processors of the computer system using the employment status module, a change to security settings for the user based on the change in the employment status of the user. The method still further comprises applying, by the one or more processors of the computer system, the change to the security settings for the user in a security management system of the threat management facility, and restricting use of at least one data processing electronic device activity on an electronic device of the user based on the change to the security settings for the user.

The above systems, devices, methods, modules, processes, and the like may be realized in hardware, software, or any combination of these suitable for a particular application. The hardware may include a general-purpose computer and/or dedicated computing device. This includes realization in one or more microprocessors, microcontrollers, embedded microcontrollers, programmable digital signal processors or other programmable devices or processing circuitry, along with internal and/or external memory. This may also, or instead, include one or more application specific integrated circuits, programmable gate arrays, programmable array logic components, or any other device or devices that may be configured to process electronic signals. It will further be appreciated that a realization of the processes or devices described above may include computer-executable code created using a structured programming language such as C, an object oriented programming language such as C++, or any other high-level or low-level programming language (including assembly languages, hardware description languages, and database programming languages and technologies) that may be stored, compiled or interpreted to run on one of the above devices, as well as

heterogeneous combinations of processors, processor architectures, or combinations of different hardware and software. In another aspect, the methods may be embodied in systems that perform the steps thereof, and may be distributed across devices in a number of ways. At the same time, processing may be distributed across devices such as the various systems described above, or all of the functionality may be integrated into a dedicated, standalone device or other hardware. In another aspect, means for performing the steps associated with the processes described above may include any of the hardware and/or software described above. All such permutations and combinations are intended to fall within the scope of the present disclosure.

Embodiments disclosed herein may include computer program products comprising computer-executable code or computer-usable code that, when executing on one or more computing devices, performs any and/or all of the steps thereof. The code may be stored in a non-transitory fashion in a computer memory, which may be a memory from which the program executes (such as random-access memory associated with a processor), or a storage device such as a disk drive, flash memory or any other optical, electromagnetic, magnetic, infrared or other device or combination of devices. In another aspect, any of the systems and methods described above may be embodied in any suitable transmission or propagation medium carrying computer-executable code and/or any inputs or outputs from same.

The elements described and depicted herein, including in flow charts and block diagrams throughout the figures, imply logical boundaries between the elements. However, according to software or hardware engineering practices, the depicted elements and the functions thereof may be implemented on machines through computer executable media having a processor capable of executing program instructions stored thereon as a monolithic software structure, as standalone software modules, or as modules that employ external routines, code, services, and so forth, or any combination of these, and all such implementations may be within the scope of the present disclosure. Examples of such machines may include, but may not be limited to, personal digital assistants, laptops, personal computers, mobile phones, other handheld computing devices, medical equipment, wired or wireless communication devices, transducers, chips, calculators, satellites, tablet PCs, electronic books, gadgets, electronic devices, devices having artificial intelligence, computing devices, networking equipment, servers, routers and the like. Furthermore, the elements depicted in the flow chart and block diagrams or any other logical component may be implemented on a machine capable of executing program instructions.

Thus, while the foregoing drawings and descriptions set forth functional aspects of the disclosed systems, no particular arrangement of software for implementing these functional aspects should be inferred from these descriptions unless explicitly stated or otherwise clear from the context. Similarly, it may be appreciated that the various steps identified and described above may be varied, and that the order of steps may be adapted to particular applications of the techniques disclosed herein. All such variations and modifications are intended to fall within the scope of this disclosure. As such, the depiction and/or description of an order for various steps should not be understood to require a particular order of execution for those steps, unless required by a particular application, or explicitly stated or otherwise clear from the context. Absent an explicit indication to the contrary, the disclosed steps may be modified,

supplemented, omitted, and/or re-ordered without departing from the scope of this disclosure.

It will be appreciated that the devices, systems, and methods described above are set forth by way of example and not of limitation. Absent an explicit indication to the contrary, the disclosed steps may be modified, supplemented, omitted, and/or re-ordered without departing from the scope of this disclosure. Numerous variations, additions, omissions, and other modifications will be apparent to one of ordinary skill in the art. In addition, the order or presentation of method steps in the description and drawings above is not intended to require this order of performing the recited steps unless a particular order is expressly required or otherwise clear from the context.

The method steps of the implementations described herein are intended to include any suitable method of causing such method steps to be performed, consistent with the patentability of the following claims, unless a different meaning is expressly provided or otherwise clear from the context. So, for example, performing the step of X includes any suitable method for causing another party such as a remote user, a remote processing resource (e.g., a server or cloud computer) or a machine to perform the step of X. Similarly, performing steps X, Y and Z may include any method of directing or controlling any combination of such other individuals or resources to perform steps X, Y and Z to obtain the benefit of such steps. Thus, method steps of the implementations described herein are intended to include any suitable method of causing one or more other parties or entities to perform the steps, consistent with the patentability of the following claims, unless a different meaning is expressly provided or otherwise clear from the context. Such parties or entities need not be under the direction or control of any other party or entity, and need not be located within a particular jurisdiction.

It should further be appreciated that the methods above are provided by way of example. Absent an explicit indication to the contrary, the disclosed steps may be modified, supplemented, omitted, and/or re-ordered without departing from the scope of this disclosure.

It will be appreciated that the methods and systems described above are set forth by way of example and not of limitation. Numerous variations, additions, omissions, and other modifications will be apparent to one of ordinary skill in the art. In addition, the order or presentation of method steps in the description and drawings above is not intended to require this order of performing the recited steps unless a particular order is expressly required or otherwise clear from the context. Thus, while particular embodiments have been shown and described, it will be apparent to those skilled in the art that various changes and modifications in form and details may be made therein without departing from the spirit and scope of this disclosure and are intended to form a part of the invention as defined by the following claims, which are to be interpreted in the broadest sense allowable by law.

The invention claimed is:

1. A method comprising:

storing, by a computer system at a threat management facility, an employment status module associated with at least one entity, wherein the employment status module associates a plurality of employment statuses with respective security settings for a plurality of users associated with the at least one entity, and the respective security settings relate to data processing electronic device activity by the plurality of users;

41

receiving, by one or more processors of the computer system, information indicating a change in an employment status of a user of the plurality of users;
determining, by the one or more processors of the computer system using the employment status module, a change to security settings for the user based on the change in the employment status of the user;
applying, by the one or more processors of the computer system, the change to the security settings for the user in a security management system of the threat management facility;
restricting, by sending instructions to a monitoring agent located on the electronic device of the user, use of at least one data processing electronic device activity of the electronic device of the user in a manner determined by a policy module of the threat management facility on an electronic device of the user based on the change to the security settings for the user; and
controlling the monitoring agent, by the one or more processors of the computer system, based on the employment status of the user and the security settings for the user, wherein the monitoring agent is configured to prevent data exfiltration based on the change of security settings for the user by preventing applications and/or websites or functionality associated with those applications or websites in a manner defined by the policy module.

2. The method of claim 1, wherein the monitoring agent is configured to monitor device usage of the electronic device and provide information to the computer system related to the monitored device usage.

3. The method of claim 2, wherein the device usage includes application usage, website usage and application installation.

4. The method of claim 1, further comprising:
receiving from the monitoring agent, by the one or more processors of the computer system, information related to device usage of the electronic device by the user;
escalating, by the one or more processors of the computer system, a user status of the user based on the received information from the monitoring agent.

5. The method of claim 4, wherein the escalating the user status of the user based on the received information from the monitoring agent comprises:
increasing the level of monitoring performed by the monitoring agent.

6. The method of claim 5, wherein the increasing the level of monitoring performed by the monitoring agent results in the monitoring agent monitoring of data flows to and from the electronic device of the user.

7. The method of claim 4, wherein the escalating the user status based on the received information from the monitoring agent comprises:
changing the security settings for the user.

8. The method of claim 4, wherein the information related to device usage of the electronic device by the user includes information indicating the user has downloaded sensitive or confidential information from a repository associated with the at least one entity.

9. The method of claim 1, wherein:
the monitoring agent is configured to monitor device usage which supports the uploading of data by the electronic device.

10. The method of claim 9, wherein preventing data exfiltration based on the change to the security settings for the user comprises:

42

blacklisting, by the monitoring agent, the use of one or more applications or websites.

11. The method of claim 9, wherein the monitoring agent is configured to monitor websites and applications that support the uploading of data from the electronic device of the user and wherein the monitoring agent is configured to alert an administrator when the user is attempting to upload information.

12. The method of claim 1, wherein the restricting the user from engaging in the at least one data processing electronic device activity on an electronic device of the user based on the change to the security settings for the user comprises:
informing, by the one or more processors of the computer system, the monitoring agent of the change to the security settings for the user.

13. The method of claim 1, wherein the monitoring agent is configured to provide information to the one or more processors of the computer system related to when the user attempts to perform data exfiltration with the electronic device.

14. The method of claim 13, further comprising:
alerting, by the one or more processors of the computer system, an administrator of the at least one entity that the user has attempted to perform data exfiltration with the electronic device.

15. The method of claim 1, wherein the employment status module maintains a list of applications and websites, and wherein the respective security settings for each of the plurality of employment statuses include settings for allowing, blocking or restricting the use of the applications and websites.

16. The method of claim 1, wherein the employment status module includes a machine learning model, the method further comprising:
predicting, by the machine learning model, a likelihood of one or more of the plurality of users to attempt to perform unauthorized data exfiltration;
based on the predicting, suggesting, by the one or more processors of the computer system, a restriction for data processing activity for the one or more of the plurality of users to an administrator of the at least one entity.

17. The method of claim 1, wherein:
the determining the change to security settings for the user based on the change in the employment status of the user,
the applying the change to the security settings for the user in a security management system of the threat management facility, and
the restricting the user from engaging in the at least one data processing electronic device activity on an electronic device of the user based on the change to the security settings for the user,
are each performed automatically and responsively to the receiving the information indicating the change in the employment status of the user.

18. The method of claim 1, wherein the threat management facility is a cloud facility.

19. A computer system, comprising:
one or more processors;
one or more memory devices coupled to the one or more processors; and
one or more computer readable storage devices coupled to the one or more processors, wherein the one or more storage devices contain program code executable by the one or more processors via the one or more memory

43

devices to implement a method for changing security settings for a user based on employment status, the method comprising:

storing, by the computer system at a threat management facility, an employment status module associated with at least one entity, wherein

the employment status module associates a plurality of employment statuses with respective security settings for a plurality of users associated with the at least one entity, and

the respective security settings relate to data processing electronic device activity by the plurality of users;

receiving, by the one or more processors of the computer system, information indicating a change in an employment status of a user of the plurality of users;

determining, by the one or more processors of the computer system using the employment status module, a change to security settings for the user based on the change in the employment status of the user;

applying, by the one or more processors of the computer system, the change to the security settings for the user in a security management system of the threat management facility;

restricting, by sending instructions to a monitoring agent located on the electronic device of the user, use of at least one data processing electronic device activity of the electronic device of the user in a manner determined by a policy module of the threat management facility on an electronic device of the user based on the change to the security settings for the user; and

controlling the monitoring agent, by the one or more processors of the computer system, based on the employment status of the user and the security settings for the user, wherein the monitoring agent is configured to prevent data exfiltration based on the change of security settings for the user by preventing applications and/or websites or functionality associated with those applications or websites in a manner defined by the policy module.

20. A computer program product for changing security settings for a user based on employment status, the computer program product comprising:

44

one or more computer readable storage media having computer readable program code collectively stored on the one or more computer readable storage media, the computer readable program code being executed by one or more processors of a computer system to cause the computer system to perform a method comprising:

storing, by the computer system at a threat management facility, an employment status module associated with at least one entity, wherein

the employment status module associate a plurality of employment statuses with respective security settings for a plurality of users associated with the at least one entity, and

the respective security settings relate to data processing electronic device activity by the plurality of users;

receiving, by the one or more processors of the computer system, information indicating a change in an employment status of a user of the plurality of users;

determining, by the one or more processors of the computer system using the employment status module, a change to security settings for the user based on the change in the employment status of the user;

applying, by the one or more processors of the computer system, the change to the security settings for the user in a security management system of the threat management facility;

restricting, by sending instructions to a monitoring agent located on the electronic device of the user, use of at least one data processing electronic device activity of the electronic device of the user in a manner determined by a policy module of the threat management facility on an electronic device of the user based on the change to the security settings for the user; and

controlling the monitoring agent, by the one or more processors of the computer system, based on the employment status of the user and the security settings for the user, wherein the monitoring agent is configured to prevent data exfiltration based on the change of security settings for the user by preventing applications and/or websites or functionality associated with those applications or websites in a manner defined by the policy module.

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